Additional results

Companion volume to *preliminary results*: deeper descriptive cuts, per-firm decompositions, and market-structure context

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This file is the companion volume to ***preliminary results****: it carries the descriptive cuts, per-firm decompositions, and regulatory / market-structure context that the headline memo only summarises. The headline methodology is the one of preliminary results — Spec A BSTS for prices and cleared MW, Spec B per-hour-class BSTS on in-band offered energy, Spec C per-curve DiD for bid shape, all with a 2024 same-calendar placebo. Where this document reports raw pre/post means or seasonality-adjusted (SA) values via a Fourier+DOW basis, those serve as descriptive sensitivity or supplementary readings, not as the identification claim — which always defers to the BSTS / DiD verdict in the preliminary memo. The pure-regression identification outputs from the earlier reform-window design live in the sibling document below.*

Contents

Glossary of acronyms

Acronym	Meaning
CCGT	Combined-cycle gas turbine: flexible, dispatchable thermal generator.
DA	Day-ahead market (Mercado Diario), cleared once per day.
IDA	Intraday auctions (3 European SIDC sessions post-2024-06-14).
MCP	Market clearing price.
MTU	Market time unit (MTU60 = hourly bidding; MTU15 = 15-minute bidding).
ISP15	Imbalance Settlement Period at 15 minutes (settlement granularity, started 2024-12-01).
REE	Red Eléctrica de España, the Spanish transmission-system operator.
OMIE	Operador del Mercado Ibérico de Energía, the wholesale market operator.
PDBF	Programa Diario Base de Funcionamiento — first post-DA viable schedule (REE-built).
PDBC	Programa Diario Base de Casación — pre-PDBF DA-cleared schedule (OMIE-built).
PHF	Programa Horario Final — post-IDA + post-RT viable schedule.
RES	Renewable Energy Sources (wind, solar PV, solar thermal, RoR hydro).
TR	Tiempo Real — real-time restrictions phase (PO 3.4–3.5), pay-as-bid.
Fase I/II	Restricciones técnicas Fase 1 (post-PDBF, pre-IDA) and Fase 2 (post-IDA), pay-as-bid.
aFRR	Automatic Frequency Restoration Reserve (Regulación secundaria), PICASSO platform.
mFRR	Manual Frequency Restoration Reserve (Regulación terciaria), MARI platform.
RR	Replacement Reserves (Reserva de Sustitución), TERRE platform.
BSP	Balance Service Provider (firm enrolled to supply balancing energy).
EUPHEMIA	Day-ahead pan-European auction algorithm (clears DA market).

Reform-window calendar and confounders

We use the same five-regime reform-window timeline as the headline memo (see *preliminary results* §1, Figure on *Full reform-window timeline*). This section adds the two calendar/confounder pieces that are too long for that memo: a calendar-coverage table that tells the reader which months each regime actually spans (and where same-calendar baselines exist), and a cross-border interconnection sanity check on whether SDAC/SIDC/XBID changes co-treat the reform windows.

Calendar coverage and same-calendar baselines

Why this matters. Spanish electricity is strongly seasonal: heating-driven demand peaks in winter, cooling in summer, hydro inflows peak in spring, solar production peaks at the summer solstice, wind blows hardest in winter. The five reform windows above do *not* span the calendar uniformly — each window covers a specific subset of calendar months, and these subsets differ across windows. Any cross-regime comparison that does not hold the calendar fixed therefore confounds the regime change with the seasonal pattern. **This applies to every variable**

in this document: bid prices, D_w, Δp / Δq, redispatch volumes, imbalance MWh, system-cost levels, RES capture rates. A reader who picks any table in this document should first read Table 1 below to know which calendar months that regime covers and what comparator is available.

Table 1: Calendar coverage of each reform window, and same-calendar baseline availability. "Calendar months covered" is the set of clock months that fall inside the window. "Days" is the window length. "Same-calendar pre-IDA placebo" indicates whether a clean same-calendar baseline exists in the pre-IDA period (2018-06-14 through 2024-06-13), where no granularity reform was active. "Reforzada overlay" flags windows in which the post-blackout reinforced-operation policy (active 2025-04-28 onward) is also in effect.

Regime	Calendar dates	Months covered	Days	Reforzada?	Same-calendar pre-IDA placebo
pre-IDA	$\leq$ 2024-06-13	all 12, multi-year	–	no	N/A (is the baseline)
3-sess	2024-06-14 – 2024-11-30	Jun–Nov	170	no	Clean (Jun–Nov 2018–2023)
ISP15-win	2024-12-01 – 2025-03-18	Dec–Mar	108	no	Clean (Dec–Mar 2018–2024)
MTU15-IDA pre-blk	2025-03-19 – 2025-04-27	late Mar – Apr	40	no	Clean (same 40 days 2018–2024)
MTU15-IDA post-blk	2025-04-28 – 2025-09-30	May – Sep	156	**yes**	Clean for granularity (May–Sep 2018–2023, pre-IDA); none for reforzada (no pre-2025-04-28 reforzada in data)
DA15/ID15	2025-10-01 – present	Oct–May (partial, ongoing)	220+	**yes**	None clean. 2024–25 same-calendar comparators are themselves a mix of post-MTU15-IDA + ISP15-win regimes

Three categories. Each regime falls into one of three buckets for the purpose of cross-regime reading:

1. **Clean same-calendar placebo available.** *3-sess, ISP15-win, MTU15-IDA pre-blk.* For each of these, we can restrict the pre-IDA baseline to the same calendar months in 2018–2024 (and average across years), and the difference (regime − pre-IDA same-cal) is interpretable as the regime effect with seasonality differenced out. Any cross-regime reading that uses these windows defaults to this same-calendar comparison unless explicitly stated otherwise.

2. **Reforzada-confounded, partial calendar-match available.** *MTU15-IDA post-blk.* The window spans May–Sep 2025, which has clean same-calendar comparators in pre-IDA (May–Sep 2018–2023). But these comparators are pre-reforzada. So the difference (MTU15-IDA post-blk − pre-IDA same-cal) confounds (a) the granularity reform, (b) operación reforzada, and (c) the post-blackout system state. There is no calendar period in the data in which reforzada is active but MTU15 is not, so reforzada cannot be differenced out via a same-calendar pre-reform window. For descriptive purposes, May–Sep 2025 is best read as "post-MTU15-IDA + reforzada + post-blackout, jointly"; see §7 on disentangling these channels.

3. **No clean calendar match.** *DA15/ID15.* The window covers Oct 2025 onward. The same calendar months one year earlier (Oct 2024 – May 2025) are themselves a mix of pre-MTU15-IDA, ISP15-win, MTU15-IDA pre-blk, and MTU15-IDA post-blk regimes — so calendar-matching to "one year ago" does not produce a single-regime comparator. The cleanest available calendar comparators for DA15/ID15 are the same months in pre-IDA years (Oct 2023 – May 2024 was largely pre-IDA, with the IDA-3 reform starting only mid-Jun 2024; Oct 2022 – May 2023 and earlier are all pre-IDA). But even with

pre-IDA same-calendar baselines, the reforzada overlay is unresolved by this comparator: DA15/ID15 is the union of MTU15-DA, MTU15-IDA, and reforzada in the same calendar window, and our data does not contain a period where any one of these three is active without the others. Therefore any DA15/ID15 reading should be flagged as descriptive only, with the three confounds explicitly named.

Fourier-deseasonalization. For a daily outcome y_t we fit a pooled Frisch-Waugh-Lovell regression on $t \in [2022\text{-}01\text{-}01, 2026\text{-}05\text{-}15]$:

$$g(y_t) \;=\; \alpha \;+\; \sum_r \beta_r \, \mathbf{1}\{\text{regime}(t) = r\} \;+\; \sum_{k=1}^{4} [a_k \, c_k(t) + b_k \, s_k(t)] \;+\; \sum_{j=1}^{6} \delta_j \, \mathbf{1}\{\text{dow}(t) = j\} \;+\; \varepsilon_t,$$

with omitted regime pre-IDA, link g equal to log for positive levels and logit for bounded shares, and day-of-year sinusoids $c_k(t) = \cos(2\pi k \ \text{doy}(t)/365)$, $s_k(t) = \sin(2\pi k \ \text{doy}(t)/365)$. Four harmonics (annual, semi-annual, four-month, quarterly) capture the seasonal cycle; the six day-of-week dummies absorb weekly periodicity. Seasonality is pooled across regimes because the 40-day MTU15-IDA pre-blk window cannot identify regime-specific Fourier coefficients.

Seasonality-adjusted (SA) value. $\hat{y}_r^{\text{SA}}$ is the predicted level of y in regime r at annual-mean Fourier (zero by orthogonality) and within-week DOW mean, with a Duan-smearing correction across in-sample residuals $\{\widehat{e}_i\}$ for the non-linear link:

$$\hat{y}_r^{\text{SA}} \;=\; \frac{1}{n} \sum_{i=1}^{n} g^{-1}\left(\widehat{\alpha} + \widehat{\beta}_r + \tfrac{1}{7} \sum_{j=1}^{6} \widehat{\delta}_j + \widehat{e}_i \right).$$

In tables, SA values sit in coloured brackets next to the raw regime mean (omitted where the value is calendar-matched by construction). Where raw and SA move the same way the pattern survives seasonality control; where they diverge, only the SA value is comparable across regimes. For the reforzada-confounded windows (MTU15-IDA post-blk, DA15/ID15) the SA value still bundles granularity, reforzada, and the post-blackout system state — it differences out the calendar, not the policy overlap.

Cross-border interconnection context

Why this matters. The day-ahead market clears through SDAC EUPHEMIA, the intraday auctions are pan-European SIDC sessions (since 2024-06-14), and the continuous intraday is XBID — so any window-by-window comparison of bid shape or clearing prices is potentially affected by changes in cross-border allocation. Three structural facts bound this channel for our reform windows.

(i) MTU15-DA was an EU-wide "Big Bang". On 2025-09-30 all SDAC bidding zones cleared a 15-min day-ahead auction simultaneously for delivery on 2025-10-01.[1] Cross-border DA allocation therefore changed identically for Spain and its neighbours at the cutover, with no Spain-specific differential. **(ii) MTU15-IDA changed the time resolution of an already-pan-European IDA auction and XBID continuous.** The 3 SIDC IDAs were European since 2024-06-14; MTU15-IDA on 2025-03-19 cut their bid time unit from 60 to 15 min, and XBID continuous moved simultaneously. The cross-border allocation *mechanism* did not change; its *time resolution* did. **(iii) Physical interconnection capacity was stable over our windows.** The 2.8 GW Spain–France link (HVDC since Oct 2015) had no reported outage in either reform window; the Bay-of-Biscay 2.0 GW expansion does not enter service until 2028.[2]

[1] SDAC 15-minute MTU Information Package (`https://www.nemo-committee.eu/assets/files/sdac-15-minute-mtu-information-paper.pdf`); OMIE press release on the go-live (`https://www.omip.pt/sites/default/files/2025-10/2501001_go_live_mtu15_md_en_vf.pdf`).

[2] Bay-of-Biscay capacity expansion plan and timeline: `https://www.pv-magazine.com/2025/06/17/france-`

Data and empirical check. ESIOS publishes 15-min cross-border balancing flows for the SEE↔FR and SEE↔PT borders at indicator IDs 535 (FR import), 536 (FR export), 539 (PT import), 540 (PT export), each in MW. Net flow per (date, clock-hour) is net = $(535 + 539) - (536 + 540)$ summed across the 4 quarters of the hour. Running the same critical-vs-flat DiD (θ on the $\mathbf{1}\{d \geq T_r\} \cdot \mathbf{1}\{h \in \mathcal{C}\}$ interaction) on net flow at each cutover:

Reform	Outcome	θ (MW)	SE	t	pre crit	post crit	Reading
MTU15-IDA	net flow	+8 965	1 751	+5.12	−9 564	−1 499	co-treated
MTU15-DA	net flow	+ 793	1 083	+0.73	+ 419	− 246	clean null

The MTU15-DA cutover is cleanly null on cross-border net flow ($t = +0.73$) — consistent with (i) the Big Bang. The MTU15-IDA cutover shows a large and significant differential (+8 965 MW), persistent across post-window truncations (dropping the last week or fortnight changes the point estimate by under 5%), so it is not a pre-blackout-stress reading; it is the direct empirical trace of (ii) finer cross-border allocation timing in the pan-European IDA and XBID venues. The DA15 cross-border channel is therefore empirically inert for our bid-shape analysis; the ID15 cross-border channel is co-treated by the same reform on a separate margin, and any ID15 bid-shape headline should be read as bundling bid-granularity with cross-border-allocation-timing within the same intervention. Source: `scripts/analysis/bid/cross_border_flow_did.py`.

Reading guide. Part A is about bidding and trading patterns. Part B is about the regulatory/post-clearing channels (reforzada, geographic dominance, CNMC sanctions, etc.) that are independent of the granularity reform but constrain how the identification can be set up.

Part I

Part A — Bidding and trading patterns

Reading guide for Part A. Three setup sections come first. §1 defines the **hour-class taxonomy** (critical / flat / midday) used to slice every table and figure that follows. §3 characterises **who price-sets** in the day-ahead market (§3.1) and, methodologically, in the intraday markets (§3.2). §4 then defines the **bid-shape metrics** in two families: *per-curve functionals* (one scalar per single bid curve — σ_p, N_{eff}) and *pairwise divergences* (a statistic of two curves — D_w, Δp, Δq). Visual results (per tech × firm × quarter bid curves with mean clearing prices overlaid) come in §4.3; quantitative tables come in §4.4.

1 Hour-class taxonomy — supplementary classifier table

The critical/flat partition is defined and motivated in *preliminary results* §2 (which uses the morning-ramp ∪ evening-ramp union as "critical", overnight hours as "flat", midday as a separate falsification class). This subsection only supplies the full per-hour classifier table that is too long for the headline memo, plus two validation checks.

`spain-increase-capacity-of-interconnection-project-from-2-8-gw-to-5-gw/`; EIB €1.6 bn financing: `https://www.eib.org/en/press/all/2025-241-eib-supports-with-eur1-6-bn-the-strategic-bay-of-biscay-electricity-interconnection-between-spain-and-france`.

Table 2: Hour-of-day classification metrics, Spain 2023–2026 (ENTSO-E A65 + A75, 15-min ISP, $n \approx 3.4$ years of daily observations per hour). The classifier is σ_{within}, the expected SD of residual demand (load − wind − solar) across the 4 quarters of each clock-hour. Critical hours: $\sigma_{\text{within}} \in [522, 1303]$. Flat hours: $\sigma_{\text{within}} \in [336, 357]$. Midday: $\sigma_{\text{within}} \in [468, 502]$ (intermediate; excluded from main specs because of the solar-saturation channel). Dropped (transition) hours sit at the regime boundaries. *Year-stability:* across 2023, 2024, 2025 the critical-set min σ_{within} exceeds the flat-set max by 1.4×; no within-class swaps across years.

Clock-hour	Demand (GW)	Solar (GW)	Wind (GW)	Residual demand (GW)	σ_{within} (MW)	CV (%)	Δ demand (MW)	Class
00:00–01:00	22.8	0.1	7.3	15.4	412	2.7	−834	*dropped*
01:00–02:00	22.0	0.1	7.1	14.8	347	2.4	−470	*flat*
02:00–03:00	21.6	0.1	6.9	14.6	336	2.3	−190	*flat*
03:00–04:00	21.5	0.1	6.7	14.7	357	2.4	176	*flat*
04:00–05:00	22.4	0.1	6.7	15.7	587	3.8	1114	*dropped*
05:00–06:00	24.4	0.2	6.7	17.5	721	4.1	1743	*critical*
06:00–07:00	26.5	1.5	6.6	18.4	915	5.0	1423	*critical*
07:00–08:00	27.7	4.9	6.2	16.6	1078	6.5	627	*critical*
08:00–09:00	28.3	9.1	5.8	13.4	1006	7.5	253	*critical*
09:00–10:00	28.3	12.0	5.5	10.9	753	6.9	−104	*dropped*
10:00–11:00	28.2	13.3	5.4	9.5	499	5.3	−66	*dropped*
11:00–12:00	28.2	13.8	5.5	8.9	468	5.3	94	*dropped*
12:00–13:00	28.3	13.8	5.7	8.8	502	5.7	−104	*dropped*
13:00–14:00	27.9	13.5	5.8	8.5	485	5.7	−331	*dropped*
14:00–15:00	27.5	12.9	6.1	8.5	469	5.5	−122	*dropped*
15:00–16:00	27.6	11.7	6.4	9.6	677	7.1	291	*dropped*
16:00–17:00	28.2	9.0	6.8	12.3	1109	9.0	553	*critical*
17:00–18:00	29.2	5.4	7.4	16.4	1303	7.9	891	*critical*
18:00–19:00	30.5	2.2	7.7	20.5	1065	5.2	884	*critical*
19:00–20:00	31.3	0.6	7.9	22.8	522	2.3	176	*critical*
20:00–21:00	30.4	0.3	7.9	22.2	622	2.8	−1444	*critical*
21:00–22:00	28.0	0.2	7.9	20.0	781	3.9	−1851	*critical*
22:00–23:00	25.8	0.2	7.7	18.0	664	3.7	−1407	*critical*
23:00–00:00	24.2	0.1	7.5	16.5	510	3.1	−1062	*dropped*

Two validation checks. (i) *No cost-shock confound*: CCGT price-setting share is 9.1% critical vs 9.0% flat — gas-cost shocks cannot propagate asymmetrically across the two classes, so within-day differencing identifies behaviour, not cost. (ii) *Outcome-side*: post-MTU15-DA the within-hour DA quarter clearing-price SD averages 6.58 EUR/MWh in critical hours vs 2.13 in flat (3.1×), so the classifier correctly predicts where post-reform quarter prices actually diverge.

2 Who clears in which hours: the tech-mix by hour-class

Before turning to who price-sets, it is worth fixing the basic descriptive picture: in each hour-class, what share of the final cleared MWh actually comes from each technology? The cross-tab in Table 3 is computed on the *final program* (PHF, post-IDA) for the post-DA15 window (2025-10-01 to 2025-12-31), so it answers the natural question "after every market has cleared, who is actually delivering the energy in each hour-class?"

Tech	Morning ramp (5–8)	Evening ramp (16–22)	Midday (11–14)	Flat (1–3)
Nuclear	**34.3**	**26.9**	25.8	**34.0**
Wind	20.9	16.3	13.7	21.6
CCGT	**14.0**	**13.2**	9.2	**13.6**
Unknown †	12.1	13.3	14.7	12.8
Hydro	10.0	13.4	4.6	9.7
Cogen	5.1	4.0	3.5	5.1
Hydro_RES	1.4	1.2	1.0	1.4
Biomass	1.2	0.9	0.9	1.2
Hydro_pump	0.9	1.9	0.1	0.5
Solar PV	0.0	8.7	**26.4**	0.0
Coal	0.1	0.1	0.1	0.1
Hour-class total (TWh)	13.9	31.1	18.5	10.5

Table 3: Final-program (PHF) share of cleared MWh by tech and hour-class, post-DA15 window. Window 2025-10-01 to 2025-12-31 (92 days). Numbers are % of total cleared MWh in each hour-class (column). PHF is the post-IDA program; we take the last-session assigned MW per (date, period, unit). The hour-class totals (TWh) reflect different hour counts per class per day (4 morning-ramp, 7 evening-ramp, 4 midday, 3 flat). † "Unknown" is the $\sim 13\%$ of cleared MWh from unit codes not in the unit-map (mostly HISVD*/GALVD*/GNVD*/GSVD* virtual-delivery families — portfolio/aggregator units that the unit-map drops); not a tech category. Reproduction: `scripts/analysis/firm/tech_mix_by_hour_class.py`.

Reading the cross-tab.

- **Nuclear is the largest single bucket in every non-midday hour-class** ($\sim 34\%$ morning ramp and flat, $\sim 27\%$ evening). Inframarginal baseload with no within-hour variation — it sets the floor of the merit order in every hour.
- **Solar PV** is the headline midday tech (26.4% midday, $\sim$0% overnight, 8.7% in the evening ramp where the ramp extends past sunset). The diurnal cycle is fully visible.
- **Wind** is the second-largest in every hour-class except midday ($\sim$20% morning and flat), dropping to $\sim$14% midday when solar takes share.
- **CCGT** (the strategic-bidder pool of *preliminary results* §4.C) clears $\sim$ 13–14% of total in every non-midday hour-class — sizeable but not dominant. Drops to $\sim$9% in midday when solar pushes it off the merit order. The strategic-engagement story of *preliminary results* §4.C is therefore about restructuring the bid-shape of a $\sim$ 13%-of-cleared-MWh fleet, not about reshuffling the merit order at any margin.
- **Cogen** (separate from CCGT, see below) clears $\sim$3–5% in every hour-class. *Its share is roughly flat across hour-classes*, consistent with the heat-driven (non- strategic) dispatch story — Cogen doesn't shift to follow the power-side demand profile because its operating decision is driven by industrial heat demand and a regulated remuneration regime, not by the wholesale price.
- **Hydro** is meaningfully higher in critical hours (evening 13.4%, morning 10%) than in midday (4.6%), consistent with the reservoir-arbitrage story (dispatch into peaks).
- **Hydro_pump** is small everywhere but skews toward critical evening (1.9% vs 0.1% midday) — discharges into the evening peak, near-zero in midday when it would be pumping.

Why CCGT and Cogen are kept separate. The unit-map distinguishes `CCGT` (58 unit codes, large combined-cycle gas turbines, dominated by the Big-4 vertically-integrated fleets) from `Cogen` (434 small cogeneration units, mostly under the regulated-renewables-and-cogeneration remuneration regime; the "$\sim$5% dispatchable tail" referenced in *preliminary results* Appendix

C.1 footnote). They are economically distinct objects: CCGT is pure power generation with a two-tier strategic bid curve and Fase I recall economics (§4.1–§7); Cogen is electricity-plus-useful-heat with heat-driven dispatch and a regulated remuneration framework that pins its incentives differently. Pooling them would dilute the strategic-engagement story (CCGT's σ_p widening in critical hours, the headline finding in *preliminary results* §4.C, would be averaged against a much larger number of small heat-driven units whose bid-shape does not move with the granularity reform). The strategic-bidding analysis throughout this document and the preliminary memo therefore restricts to CCGT.

What the table is and is not. This is a descriptive post-DA15 snapshot, not a treatment effect. It tells the reader which technology buckets are sizeable in which hour-class, which contextualises every subsequent table in this document (a 3% share moving by 30% is much less of a system event than a 14% share moving by 30%, even if the percentage-point change is the same). For pre-vs-post comparisons we use the BSTS / DiD machinery of *preliminary results* §3.A–C; this table is not a reform-effect estimate.

3 Who price-sets

3.1 For the day-ahead market

Method. OMIE submits stepwise orders to EUPHEMIA: in-the-money fully accepted, out-of-the-money rejected, **at-the-money curtailable**. The price-setter is the unit whose at-MCP step is *partially* accepted: with q_{below} the MW bid strictly below MCP, q_{at} the MW at MCP ($\pm$0.01 EUR/MWh), and q_{assigned} the unit's cleared MWh,

$$q_{\text{below}} < q_{\text{assigned}} < q_{\text{below}} + q_{\text{at}}.$$

Full acceptance (inframarginal) and curtailed-to-zero (often a minimum-income deactivation) are excluded. A naive rule flagging every $p_{\text{bid}} = \text{MCP}$ — the CNMC market-supervision metric — pools all three cases and is mechanically larger: for CCGT DA15/ID15 critical it gives 9.1% against our 2.7% (most CCGT at-MCP steps are minimum-income-deactivated to zero). Aggregations are MW-weighted by q_{at}.

Who is marginal. Across the five reform regimes, **wind is the largest sell-side price-setter** in almost every cell (18–71% of at-MCP MW): renewable-aggregator portfolios systematically land a single block at-the-money. Hydro and pump-storage are next; in DA15/ID15 Iberdrola owns the largest share of marginal MW ($\approx$ 65% critical, 80% flat) through pump-storage, hydro and its wind aggregator. **CCGT's share is small and volatile** ($\sim$1–17% in critical hours; a lone 33% flat-hour spike under DA60/ID15 post-blk is 80% Endesa). One cross-check worth keeping: CCGT's critical-versus-flat shares are near-symmetric in 3-sess (8.0/8.9) and DA15/ID15 (2.7/2.0), so a TTF/gas-cost shock — which hits both hour-classes alike — is not what the within-day differencing detects. Within the CCGT pool, the per-firm split:

Regime	Hour-class	IB	GE	GN	HC	OTHER
3-sess	Critical	**60**	31	0	0	8
3-sess	Flat	16	**50**	0	0	25
ISP15-win	Critical	35	**57**	0	0	5
ISP15-win	Flat	2	**74**	0	3	20
DA60/ID15 post-blk	Critical	31	**69**	0	0	0
DA60/ID15 post-blk	Flat	19	**80**	0	0	0
DA15/ID15	Critical	**65**	29	0	6	0
DA15/ID15	Flat	**80**	0	6	14	0

Table 4: Per-firm shares within the CCGT price-setter pool (% of CCGT q_{at}). **GN's CCGT bid stack is so heavily concentrated far above the clearing-price ceiling (a flat shelf at** $\sim$1,000 **EUR/MWh) that GN essentially never lands at MCP**. IB and GE alternate as the dominant CCGT price-setter. The 33.1% CCGT flat-hour figure in DA60/ID15 post-blackout is 80% GE.

Price-setting is not market power, for CCGT. The partial-acceptance metric identifies the *marginal unit* — whoever's at-MCP step is curtailed to clear the market. For price-taking technologies (wind, hydro) that fairly describes who sets the price; for CCGT it is *not* a conduct measure, and the table above is best read as a warning. A unit is flagged merely for having offered MW at the price that happened to clear: a thin, near-inelastic block at MCP makes it mechanically marginal with no exercise of power. The ranking even *inverts* the market-power ordering — GN, which withholds the most (75% of its CCGT MW parked at scarcity prices; the two-tier bid curve, §4.1), scores 0% precisely *because* it bids too high ever to be marginal, while IB tops the CCGT pool only by keeping a thin competing tier near MCP. CCGT market power is not exercised *inside* the day-ahead auction; it is exercised by withholding *from* it and captured downstream in pre-IDA Fase I redispatch (§7), which dominates every Big-4 CCGT fleet's dispatched energy. For CCGT we therefore read conduct from the bid-vs-marginal-cost evidence (§4.1) and the Fase I channel, not from the price-setter share.

3.2 For the intraday markets

The same partial-acceptance rule, applied to the intraday-auction clearing prices, bid stack and programs (sell side, pooled across the 3 IDA sessions per day), shows **CCGT far more active as an IDA price-setter than as a DA one**: 24.2% of critical-hour IDA mass under DA15/ID15 against 2.7% in DA — a factor of ~ 9. This is the price-setter-side trace of the day-ahead-to-intraday substitution documented in §4.1: CCGTs bid above the day-ahead clearing price but compete in the intraday auction, where they clear at the marginal step. Wind still dominates ($\sim 32\%$ critical) but by a smaller margin than in DA.

4 Bid-curve shape: per-curve functionals and pairwise divergences

We measure bid-curve shape with two conceptually distinct families of statistics, and keep them separate throughout.

- **Family 1 — per-curve functionals.** A functional maps a *single* bid curve to a scalar, $T : f \mapsto T(f) \in \mathbb{R}$; no comparison between curves enters its definition. Cross-regime, cross-firm and cross-hour reading is then done by comparing the *values* $T(f)$ across curves. This family holds the price spread σ_p and the effective tranche count N_{eff} (§4.1). **This is the primary descriptive object**: it asks what one bid curve looks like, and how that value moves across regimes.
- **Family 2 — pairwise divergences.** A divergence is a statistic of *two* curves, $D(f_i, f_j)$: it does not summarise either curve on its own, it measures how far apart they are. This family

holds D_w and its Δp / Δq channel decomposition (§4.2). We use it for one comparison only — a unit's four within-hour quarter curves against each other — as a within-hour-shaping diagnostic, secondary to Family 1.

The distinction is the one the advisor asked for: characterise each bid curve by its own value (Family 1), rather than by divergences between curves (Family 2). Family 2 is retained only because the within-hour quarter comparison has no single-curve analogue.

4.1 Family 1 — per-curve functionals: σ_p and N_{eff}

The strategic band, chosen data-driven. Both families restrict to a price band $|p - \text{MCP}| \le h$ — the region where a bid is in contention. We read h off the data. Figure 1 plots the MW-weighted density of the bid-to-clearing distance $|p_{\text{bid}} - \text{MCP}|$ over all day-ahead sell tranches of the price-setting techs; it is **bimodal** — a dense near-MCP *competing* cluster and a sparse far *scarcity-cap* cluster — and the competing cluster ends in a sharp cliff at $h = 140$ EUR/MWh. This is corroborated decisively by the price side: the day-ahead clearing price has a median of 65, exceeds 205 EUR/MWh in only 0.1% of periods, and *never* exceeds 250 over the whole 2024–26 sample — so median MCP $+h$ sits exactly at the realised price ceiling. A bid more than $\sim$140 above the clearing price essentially cannot clear: it is withholding, not competition, and $\sim$200 EUR/MWh is the absolute price above which a bid clears in 0.1% of periods. (An earlier draft set $h = 230$ at the density antimode — the valley floor beyond the cluster — which reached MCP $+ 230 \approx 295$, above the all-time-maximum clearing price.) h is estimated once, pooled across regimes and techs, and frozen.

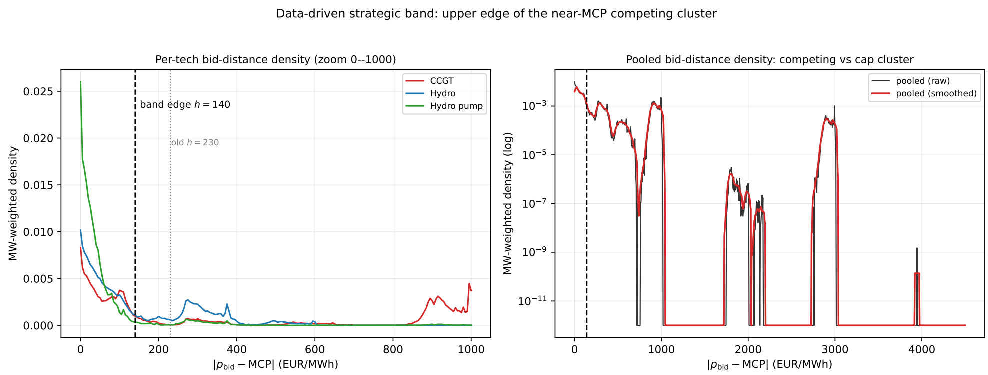


Figure 1: Data-driven strategic band. MW-weighted density of $|p_{\text{bid}} - \text{MCP}|$ for day-ahead sell tranches (price-setting techs, all regimes pooled). Left: per-tech, zoomed to 0–1000 EUR/MWh. Right: pooled, log scale — the near-MCP competing cluster and the scarcity-cap cluster show as two modes. The band edge $h = 140$ (dashed) is the upper edge of the competing cluster; it coincides with the clearing-price ceiling (median MCP $+140 \approx$ the 99.9th percentile of the day-ahead price). The earlier $h = 230$ (dotted) sat in the valley, above the realised price ceiling.

The two-tier bid curve, read against marginal cost. The bimodality of Figure 1 is, unit by unit, a **two-tier bid curve**. Figure 2 aggregates each Big-4 CCGT fleet's day-ahead sell tranches into one supply curve and reads it against the bottom-up CCGT marginal cost (grey band, $\sim$100–110 EUR/MWh in DA15/ID15; §7.5) — the competitive benchmark. A low *competing tier* is offered around marginal cost or below and clears the auction; a high *scarcity tier* is parked far above any plausible clearing price — it does not clear the auction — it is *deliberate withholding*. The parked capacity is what REE recalls in the pre-IDA Fase I technical-restrictions process (§7); that recall is remunerated pay-as-bid, but through a *separate* restriction offer the unit submits to REE — a distinct instrument from the day-ahead bid. The day-ahead scarcity-tier price therefore carries no payment of its own; its level is a firm-specific focal choice.[3] The withholding boundary is the $\sim$200 EUR/MWh clearing-price ceiling: a bid above it essentially cannot clear. The split is sharp and firm-specific. Naturgy is the extreme withholder: a thin sliver below the clearing price, then 75% of its MW parked in a flat shelf at $\sim$1,000 EUR/MWh. But it is not a Naturgy-only story — *every* Big-4 fleet withholds a comparable share of the rest: Endesa 22% above 200 (much at the 4,000 cap), **Iberdrola** 22%**, EDP-HC** 19%. Iberdrola has no extreme cap-shelf, so it looks competitive against a 500 EUR/MWh cutoff — but against the true $\sim$200 ceiling it withholds as much of its CCGT MW as Endesa does, in a 200–400 ramp, and clears only $\sim$16% of its final programme in the day-ahead (barely above Naturgy's $\sim$14%). The firms differ in *how far* they bid above the ceiling — Naturgy blatantly, the others quietly — not in *whether* they withhold. *This bid-versus-cost picture, not the price-setter share (§3.1), is the conduct measure for CCGT*: the per-curve functionals are computed on the competing tier, where price-graded bidding is a genuine choice; the scarcity tier is a separate withholding margin, excluded by the band.[4]

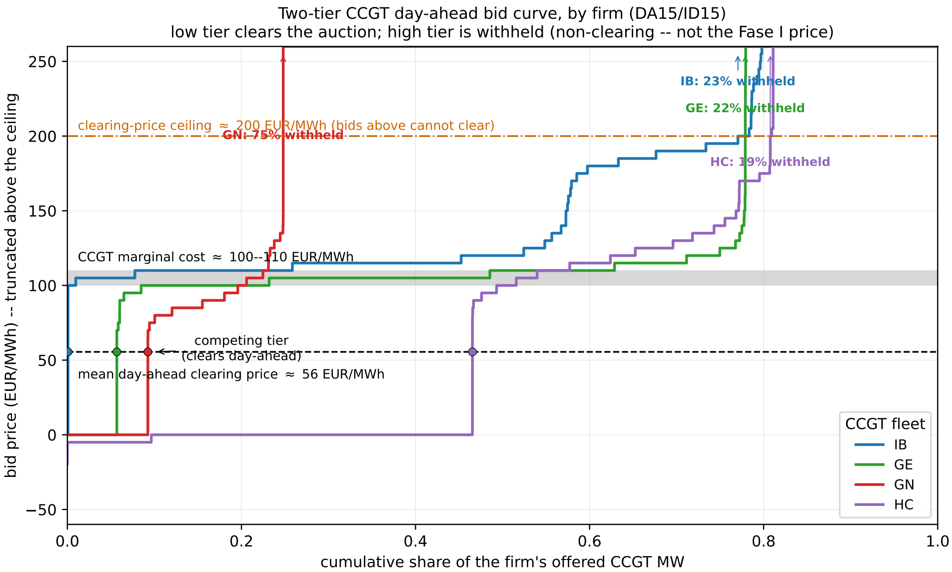


Figure 2: Two-tier CCGT day-ahead bid curve, by firm (DA15/ID15). Each Big-4 CCGT fleet's day-ahead sell tranches aggregated into a supply curve: bid price vs cumulative share of offered MW. Dashed line: mean clearing price; grey band: bottom-up CCGT marginal cost; dash-dot line: the $\sim$ 200 EUR/MWh withholding threshold (the day-ahead price exceeds it in 0.1% of periods). *Y-axis truncated just above the ceiling*: the parked scarcity-tier prices (Naturgy $\sim$ 1,000, Endesa at the 4,000 cap, Iberdrola/EDP-HC on a 200–700 ramp) are non-clearing markers, not the Fase I payment (Fase I is paid pay-as-bid through a separate restriction offer, see footnote), so plotting them at face value would invite a misreading. The economically meaningful information is the cliff at the ceiling and the withheld share annotated on each firm's curve — Naturgy 75% withheld, the others 19–23%. All four fleets clear only a small minority of their programme in the day-ahead.

[3] Process: P.O. 3.2, Restricciones técnicas (Resolución CNMC de 6-mar-2025, `BOE-A-2025-5342`). Settlement: P.O. 14.4, apartado 20 (Resolución CNMC de 12-jun-2025, `BOE-A-2025-13076`): apartado 20.1 — Fase I up-redispatch with a restriction offer is paid pay-as-bid *that* offer (the *oferta de restricciones técnicas*, submitted separately to REE); apartado 20.2 — the no-offer default is 1.15× the day-ahead price. Per-unit restriction offers are not published, so the Fase I rent is read from the aggregate Fase I price against marginal cost (§7.5), not from observed bids. Bidding high in the day-ahead auction to be excluded and then dispatched in restrictions is the conduct CNMC sanctioned in 2019 (`SNC/DE/174/17`, `SNC/DE/175/17`).

[4] Independent corroboration: Brown and Reguant (Univ. Alberta WP 2026-04, "The Price Impacts of Renewable Power: A Tale of Two Sources") document HHI > 4,000 in the day-ahead restrictions market and a median of just 4 firms with accepted upward bids on the 2016–2024 sample, matching the Big-4 CCGT scarcity-tier story characterised here at the bid level.

The intraday auctions show it is strategy, not cost. The pre-IDA Fase I redispatch settles the CCGT grid-security role *before* the intraday auctions clear (post-IDA RT on CCGT is negligible until DA15/ID15; §7), so the intraday auction is a market in which the Fase I-recall game is no longer in play. Figure 3 puts each Big-4 CCGT fleet's day-ahead and intraday-auction supply curves side by side, and the contrast is total: in the day-ahead the curve is two-tier, with 19–75% of MW withheld above the ∼200 ceiling; in the intraday auction the *same fleets, same days* bid a single competitive tier around marginal cost — a MW-weighted 82–96 EUR/MWh and ∼0% withheld, for all four firms, Naturgy included (90 against its 767 day-ahead). A unit's marginal cost does not change between two auctions held hours apart; the firm simply reveals it in the intraday one. **The day-ahead bid is therefore not a cost — the scarcity tier is priced purely *not to clear*, holding the capacity back for the pre-IDA Fase I recall, which is paid through the separate restriction offer.** This is the cleanest evidence in the document that the day-ahead two-tier curve is withholding rather than cost-reflective bidding.

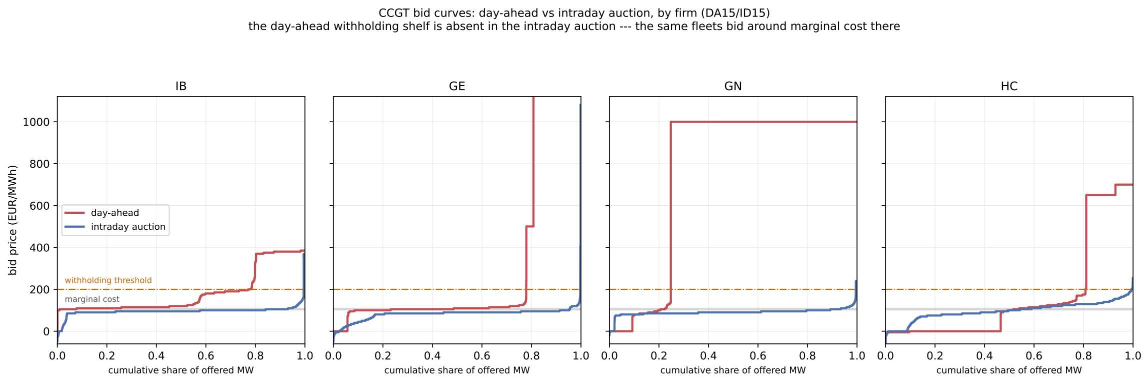


Figure 3: CCGT bid curves: day-ahead vs intraday auction, by firm (DA15/ID15). Each Big-4 CCGT fleet's sell tranches aggregated into a supply curve (bid price vs cumulative share of offered MW). Grey band: bottom-up CCGT marginal cost; dash-dot line: the ∼200 EUR/MWh withholding threshold. The day-ahead curve (red) carries the withholding shelf; the intraday-auction curve (blue) sits on the marginal-cost band for the whole book. Same units, same days — the withholding is a day-ahead-only strategy.

Per-curve functionals — definitions in the headline memo. The two per-curve scalars σ_p (MW-weighted SD of in-band tranche prices) and N_{eff} (inverse Herfindahl of in-band tranche MW shares) are defined in *preliminary results* §1 and used as the bid-shape outcomes of Spec C (*preliminary results* §3.C, 4.C). We use the same definitions here without re-deriving them, and the same data-driven band $|p - \text{MCP}| \leq h$ as in the strategic-band paragraph above. The functionals decouple cleanly: a unit can split into many tranches at nearly one price (high N_{eff}, low σ_p) or place two blocks far apart (low N_{eff}, high σ_p).

Offer types: the warning for CCGT. The three OMIE offer types (simple price-quantity ladder; minimum-income condition (MIC); block order) are described in *preliminary results* §1 footnote and the composition shift across regimes is analysed in *preliminary results* Appendix A.7. The take-away for the per-curve functionals: simple offers are read at face value; block orders mechanically force $\sigma_p \to 0$ and $N_{\text{eff}} \to 1$ (one price held across many periods, no within-period structure); MIC offers displace some shaping onto the income floor. The mix is sharply tech-specific (DA15/ID15 sell offers): wind / solar / hydro / pump-storage are ∼100% simple; nuclear is 64% simple; **CCGT is** 18% **simple,** 50% **MIC,** 32% **block** — the functionals are clean for renewables and hydro, but for CCGT any geometry-to-conduct step must condition on offer type.

4.2 Family 2 — pairwise divergences: D_w, Δp, Δq

D_w, in full. For a cell (firm, unit, date, clock-hour) the hour splits into 4 quarters. In quarter q the unit's band-restricted sell curve is the right-continuous step function $S_q(p) = \sum_k x_{qk}\,\mathbf{1}\{p_{qk} \le p\}$ on $p \in [c-h, c+h]$, with $S_q(c-h) = 0$ and $S_q(c+h) = M_q$ (the band-mass), where c is the hour-mean of the four quarter clearing prices — one reference common to all 6 quarter-pairs, so the band does not move across quarters. For two quarters i, j the kernel-weighted Wasserstein-1 dissimilarity is

$$D_w^{(i,j)} \;=\; \int_{c-h}^{c+h} K_h(p-c)\,\big|S_i(p) - S_j(p)\big|\,dp, \qquad K_h(u) = \max\big\{0,\ 1-(u/h)^2\big\}\,,$$

with K_h the Epanechnikov kernel down-weighting bids far from clearing; D_w has units EUR. The cell statistic is the mean over the 6 quarter-pairs (for the binary flag $D_w > 0$, mean, max and "any pair" agree).

Price and quantity channels. Splitting the underlying Wasserstein-1 distance at the common mass $\min(M_i, M_j)$ separates a price channel from a quantity channel,

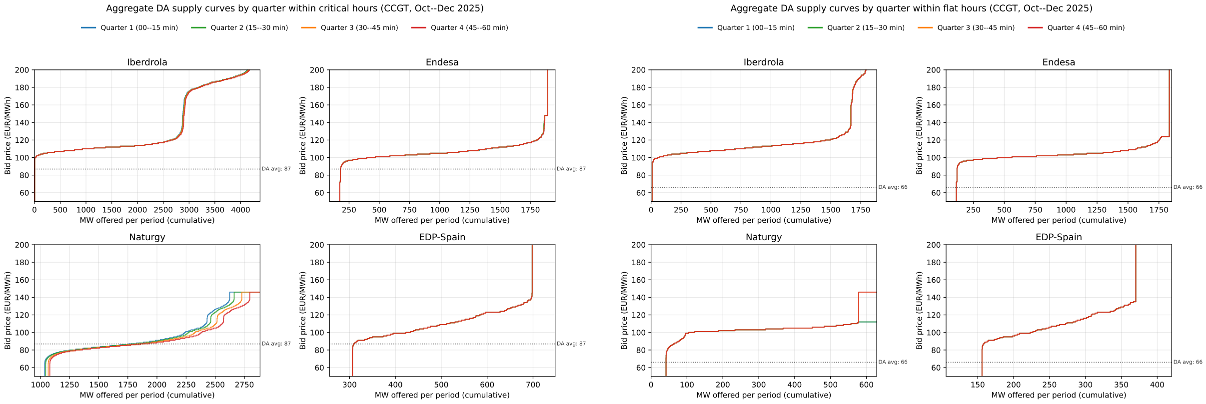
$$\Delta q_{ij} = |M_i - M_j| \ \ [\text{MWh}], \qquad \Delta p_{ij} = \frac{1}{\min(M_i, M_j)} \int_0^{\min(M_i,M_j)} \big|S_i^{-1}(Q) - S_j^{-1}(Q)\big|\,dQ \ \ [\text{EUR/MWh}].$$

Δp and Δq are read as channel *indicators* (active/inactive, and which dominates), not additive contributions — raw Δq in MWh is scale-dependent, so we lean on the active/inactive distinction.

Band, and a pending refresh. D_w, Δp and Δq use the same data-driven band as Family 1 ($h = 140$, §4.1). *Pending:* the D_w / Δp / Δq tables in §4.4 are still at the legacy $h = 50$ — a $\sim 4.6\times$ wider band can saturate the binary $D_w > 0$ flag, an analytical change held as a separate step rather than a mechanical re-run.

4.3 Bid graphs by tech, firm, and within-hour quarter

The figures below show the aggregate within-hour quarter bid curves with the mean clearing price overlaid (so the strategic band $|p - c| \le 140$ is visible). Aggregation across $\sim$ 92 DA15/ID15 dates per (firm, hour, quarter) cell smooths per-day shaping — the per-curve functionals of §4.4 are the systematic measure, these are the visual companion. *Reading*: same-quarter overlap in flat hours is the falsification; visible quarter spread in critical hours is the strategic signal, concentrated in GN CCGT and Endesa Hydro. The IDA-side and pump-storage-demand analogues are similar and not reproduced.



(a) Critical hours (05:00–09:00, 16:00–23:00) **(b)** Flat hours (01:00–04:00) — falsification

Figure 4: Aggregate DA supply curves by within-hour quarter, CCGT per firm, Oct–Dec 2025. Quarter overlap in flat hours is essentially complete (the falsification); critical-hour quarter spread is visible for GN.

4.4 Results: per-curve functionals, and the pairwise D_w diagnostic

The results below quantify what the bid-curve figures in §4.3 show visually: first the Family-1 per-curve functionals (σ_p, N_{eff}, defined in §4.1), then the Family-2 pairwise D_w diagnostic (§4.2).

Per-curve functionals: price spread and effective tranche count. Table 5 and Figure 5 report σ_p and N_{eff} averaged over (unit, date, period) day-ahead sell-bid curves, by tech, hour-class and regime — the cross-regime reading is a comparison of per-curve *values*, with no differencing of curves.

Table 5: Per-curve bid-shape metrics by tech, hour-class and regime (DA sell bids, in-band $|p - \text{MCP}| \leq 140$ — the data-driven band of §4.1 — mean over (unit, date, period) curves). σ_p = MW-weighted price spread (EUR/MWh); N_{eff} = effective in-band tranche count. 953,384 in-band curves.

		3-sess		ISP15-win		DA60/ID15 pre		DA60/ID15 post		DA15/ID15	
Tech	Hour	σ_p	N_{eff}	σ_p	N_{eff}	σ_p	N_{eff}	σ_p	N_{eff}	σ_p	N_{eff}
CCGT	Critical	36.1	3.3	32.0	4.2	29.2	3.1	24.5	3.1	24.8	3.3
CCGT	Flat	27.5	2.5	45.2	2.7	17.1	2.3	20.2	2.6	18.2	2.3
Hydro	Critical	17.5	4.0	16.3	4.0	13.1	3.5	15.5	3.5	16.6	3.7
Hydro	Flat	17.6	4.0	18.5	3.8	12.6	3.5	16.0	3.5	15.1	3.8
Hydro pump	Critical	3.6	2.5	3.8	2.5	2.6	2.3	3.9	2.8	3.8	3.4
Hydro pump	Flat	2.1	2.6	3.9	2.9	2.4	2.3	2.6	2.4	2.8	4.1

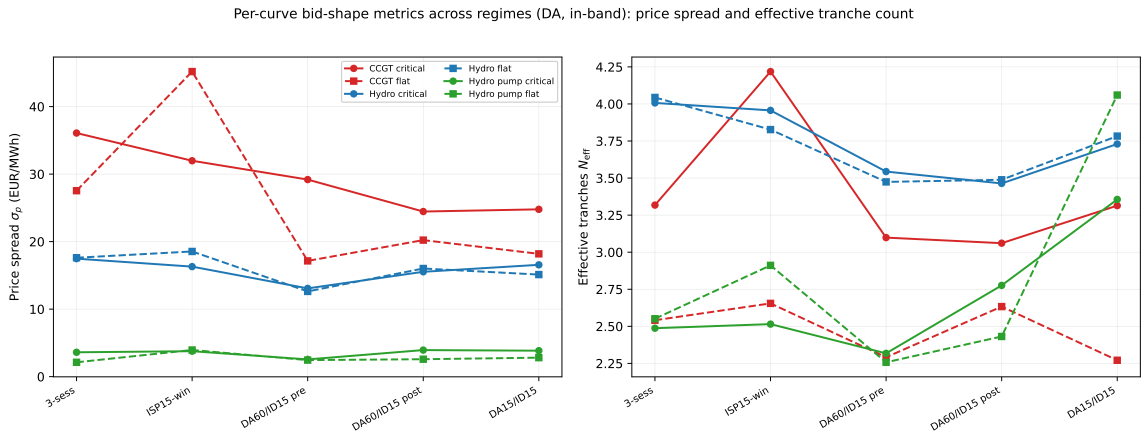


Figure 5: Per-curve bid-shape metrics across regimes (DA, in-band, data-driven band $|p - \mathrm{MCP}| \leq 140$). Left: price spread σ_p; right: effective tranche count N_{eff}. Each line is a (tech, hour-class) cell; solid = critical hours, dashed = flat. CCGT carries the widest price spread ($\sigma_p \approx$ 25–40 EUR/MWh) and it is larger in critical than flat hours in four of the five regimes — the strategic signal; Hydro is intermediate ($\sim$ 13–18) and Hydro pump near-flat ($\sim$ 3–4). N_{eff} sits at $\sim$ 3–4 for the price-setting techs.

The pairwise D_w diagnostic (Family 2). D_w asks whether a unit's four within-hour quarter curves *differ* near clearing; the per-cell binary flag $D_w > 0$ is the within-hour-shaping signal. The cleanest reform-window test is the IDA blackout split for CCGT — pre-blackout (2025-03-19→04-27, MTU15-IDA active, reforzada-free) vs post-blackout (04-28→09-30, reforzada). If REE post-clearing intervention removed the leverage on the clearing price, pre > post.

Table 6: IDA quarter-dissimilarity, pre vs post-blackout, CCGT. Percentage of (firm, unit, date, hour) cells with $D_w > 0$ (i.e., quarters differ within the Epanechnikov band, $h = 50$ EUR/MWh, around hour-mean IDA clearing price). *Pre-blackout*: 2025-03-19 → 2025-04-27 (40 days, MTU15-IDA active, reforzada-free). *Post-blackout*: 2025-04-28 → 2025-09-30 (155 days, reforzada active).

	Pre-blackout			Post-blackout		
Firm	Critical	Flat	Crit−Flat	Critical	Flat	Crit−Flat
IB	9.6	7.0	+2.6	20.3	6.2	+14.1
GE	14.1	10.0	+4.1	43.8	33.2	+10.6
GN	30.5	9.2	**+21.3**	53.8	37.9	+15.9
HC	27.9	20.7	+7.2	34.2	21.2	+13.0

Reading. The directional prediction (pre > post) is *rejected* — $D_w > 0$ rates roughly double post-blackout for every firm. But the critical-vs-flat asymmetry survives and broadens: pre-blackout only GN shows a clean gap (+21pp), post-blackout all four firms do (+11 to +16pp). The strategic within-hour mechanism is essentially unchanged from pre-reforzada; the relevant new lever is geographic (§8). A complementary $\Delta p/\Delta q$ split (DA, CCGT) shows GN shaping is quantity-only (18% of cells Δq-active, $\sim$ 0% Δp-active) — the Cournot version of price-raising: a fixed near-cap price ladder with quarter-to-quarter quantity reallocation.

Within-hour shaping by firm and unit. A per-unit $\Delta p/\Delta q$-activity scan of DA15/ID15 critical hours confirms within-hour shaping is exploited by specific firm-unit combinations, not uniformly. GN never price-sets at MCP (its bids are too high, §3.1) yet shapes within-hour bids in 33% of critical hours with the largest median Δp of any CCGT firm; Iberdrola is the most active shaper across CCGT (68% of cells Δp-active), hydro (74%) and pump-storage (88%); among wind aggregators Gesternova ladder-bids across quarters (96% Δp-active) while AXPO and NEXUS bid identically across all four.

Wind within-hour variation is forecast, not strategy. A natural worry is that the quarter-to-quarter movement in wind bids is strategic scarcity-chasing rather than the wind forecast. We test it on the functional price metric itself — regressing each wind curve's σ_p on a within-hour scarcity proxy (the quarter's day-ahead load forecast net of solar, i.e. residual demand *excluding* wind and so exogenous to any wind unit, deviated from its clock-hour mean) with unit×date×hour fixed effects, so only across-quarter within-hour variation identifies the slope (DA15/ID15, 2.3M in-band wind curves, date-clustered SEs). The band is centred on the *clock-hour-mean* clearing price rather than each quarter's own, so within-hour σ_p movement is ladder shaping and not the band sliding with the quarter's clearing price. Four facts: (i) 84% of wind in-band curves are a single block ($\sigma_p = 0$) and 89% of clock-hours show no across-quarter σ_p variation; (ii) where σ_p does move, the scarcity response is statistically positive but economically negligible — +0.01 EUR/MWh per GW pooled (within-hour partial correlation +0.01), +0.09 on the σ_p-varying subsample; (iii) it holds hour by hour — critical hours give the same slope (+0.009) and no clock-hour of the day reaches $|\beta| = 0.06$; (iv) the largest response, Gesternova's ($\beta = +0.14$, partial correlation +0.09), is still tiny — the small aggregator that ladder-bids across quarters — against $\beta \approx 0$ for the other large aggregators (AXPO, NEXUS, IGNIS). The quantity functional N_{eff} and the bid-price level give the same null, and the result is robust to controlling for the unit's own within-hour forecast (offered-MW) shape and to plain load. Within-hour wind movement is the sub-hourly forecast; the functionals do not respond to scarcity — so the wind functionals (§4.1) read as a price-taking baseline.

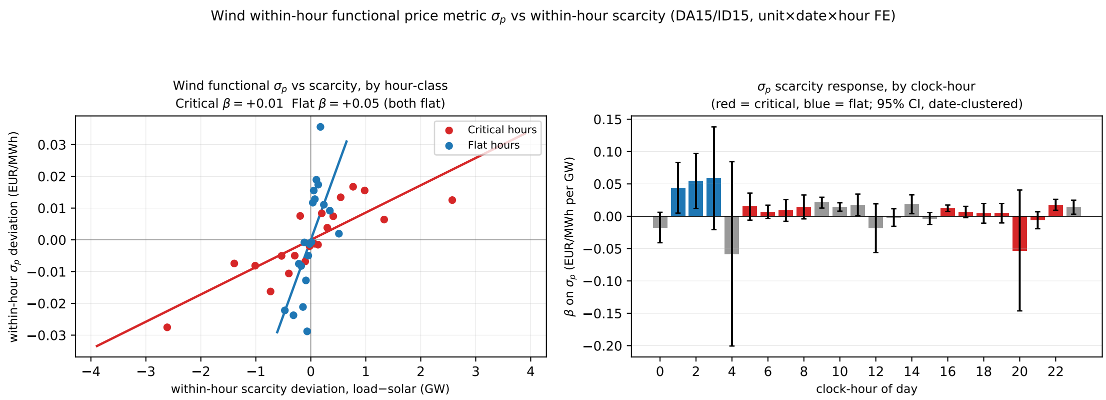


Figure 6: The wind functional price metric σ_p does not respond to within-hour scarcity (DA15/ID15, unit×date×hour fixed effects; band centred on the clock-hour-mean clearing price). Left: binned within-hour-demeaned σ_p against the within-hour-demeaned scarcity proxy (load minus solar forecast, GW), for critical and flat hours — both slopes essentially flat. Right: the slope for each clock-hour of the day (red = critical, blue = flat hours; date-clustered 95% intervals) — no hour reaches an economically meaningful response.

Pump-storage within-day arbitrage under MTU15-DA: a descriptive look (no causal claim). The bid-shape evidence above is one side of the granular-pricing reform; one would naturally look at pump-storage for the quantity-side mirror. Pre/post means on the two sides of the pump-storage book, taken at face value, show a striking within-day reallocation:

	Generation (MW per period)		Pumping (MW per period)	
Hour-class	Pre (Jul–Sep)	Post (Oct–Dec)	Pre	Post
Critical	851	1 018 (+20%)	664	317 (−52%)
Flat	724	258 (−64%)	52	566 (+10×)

Generation appears to reallocate from flat hours (−64%) to critical hours (+20%); pumping mirrors it. Total daily generation is essentially unchanged (14 874 → 14 250 MWh, −4%). The reading would be textbook within-day arbitrage: pre-MTU15-DA, hourly bids could not price-discriminate across quarters; post-reform, the unit class with the largest within-hour arbitrage value exploits within-hour price spreads cleanly.

The headline-methodology verdict is more cautious. The BSTS counterfactual on per-tech cleared MWh (Spec A in *preliminary results* §4.A) does not retain the pump-storage MTU15-DA effect under the extrapolation-drift diagnostic (see *preliminary results* §4.A.6, "BSTS extrapolation drift cases"). The pre/post raw means above remain descriptively informative — the within-day reallocation pattern is real in the data — but we do not interpret them as a causal MTU15-DA effect for the reasons set out in the preliminary memo (cross-season pre/ post, drift between the 14-day and 40-day post-windows, sensitivity to the choice of pre-window).

5 Continuous-intraday repositioning

The bidding-side mirror of the cost cascade (§9) is the continuous intraday market (XBID): if a finer-grained auction absorbs within-hour scarcity, XBID repositioning volume should fall. The per-tech, per-firm and per-regime breakdowns reduce to four facts.

- **Volume falls, then partly recovers.** Deseasonalized gross XBID activity is ∼ 32 → 12 → 23 GWh/day across the five regimes (−29% baseline to DA15/ID15, trough −62% in the 40-day MTU15-IDA-only window); CI's share of OMIE-cleared volume traces the same path (∼ 3% → 1.7% → 2.9%). The granular auction absorbs scarcity that previously fell to XBID and real-time redispatch.
- **Bilateral contracts dominate throughout** (∼ 60% of dispatched volume), with renewables' bilateral share rising under DA15/ID15 (Wind 20 → 43%, Solar PV 15 → 33%).
- **CCGT is a systematic net seller** in XBID in every regime (+0.8 to +2.5 GWh/day) — the continuous-market face of "stay out of the day-ahead"; nuclear a systematic net buyer (−1.7 to −2.9 GWh/day post-blackout). CCGT and Hydro sell at a small premium to the same-day day-ahead price (∼+6/MWh); Nuclear and Solar PV at a large discount (−20 to −28). The cross-tech price structure is stable; only volumes move.
- **Within-hour rebalancing intensity is unchanged.** Per-firm dispersion of XBID sell volume across the four quarters of a clock-hour is 2–5× larger in critical than flat hours — the same critical-flat asymmetry as the auction-side evidence, ranked IB > GN > HC > GE — and is stable across the post-MTU15-IDA regimes. CI *volume* fell with the granular auction; per-firm within-hour rebalancing *intensity* did not.

Part II
Part B — Regulatory and confounder context

The REE program-and-intervention chain (brief primer). Every delivery day a firm program for each unit-period is built up through a chain of OMIE clearings and REE redispatch steps. Each step writes the next-level program (PDBC → PDBF → PDVP → PHF → PHFC → P48); each transition involves either a market clearing or a REE technical-restriction (RT) intervention. The schematic is:

Step	New program	Mechanism
DA clearing (D-1, 12:00)	**PDBC**	OMIE marginal-price DA cleared MW
+ bilateral contracts	**PDBF**	OMIE adds BRP-communicated bilaterals
+ REE **pre-IDA RT**	**PDVP**	*Fase I* (pay-as-bid congestion + reserve) + *Fase II* (offer redispatch)
+ IDA session s+ REE **post-IDA RT**	**PHF**(s)	Per-session IDA clear + REE security tweak ("RT2"); 3 sessions in SII
+ continuous round r+ REE post-continuous RT	**PHFC**(r)	24 rounds/day; per-round continuous + REE tweak
+ real-time RT (delivery, D)	**P48**	REE *Tiempo Real* (TR) intervention; live updates

REE's redispatch energy comes from balancing services activated alongside — FCR (primary), aFRR (secondary), mFRR (tertiary), RR, Gestión de Desvíos, RPA — which together close the PDBF-to-delivery gap; imbalance settles per ISP, dual-priced. Two recurring labels: *Reforzada* = REE's continuous post-blackout (2025-04-28 onward) stance of larger pre- and post-IDA RT, mostly Fase I-up; *RT2* = the post-IDA RT leg, measured as PHF − PIBCA at maximum-session rows. Both are confounders for the MTU15-DA comparison (its pre- and post-windows are both post-blackout). *Part B roadmap*: the phenomenon (§6, the CCGT q_1 drop), the absorbing mechanism (§7, REE post-clearing inclusion), the enabling structural feature (§8, geographic CCGT concentration), the system-cost view (§9), and the enforcement context (§10).

6 The q_1 drop: CCGT auction-cleared collapse

Variable. q_1 = sum of PDBC (OMIE DA auction-cleared) MWh, restricted to the Big-4 pivotal fleet (*Iberdrola, Endesa, Naturgy, EDP-HC*), aggregated per (technology, calendar month). Programs used: PDBC (OMIE), PDBF (PDBC + bilateral contracts, OMIE), PHF (post-IDA program at max session, OMIE). **Same-calendar-month** comparison absorbs within-year seasonality.

6.1 The three-component decomposition

The Big-4 fleet's monthly volume per technology decomposes as q_{final} = PDBC + Bilaterals + (IDA + REE-RT). The figures below separate the three components weekly for the Big-4 aggregate and for CCGT by firm: CCGT's DA-cleared layer collapses post-blackout while the IDA + RT layer balloons, and the shift is firm-wide — roughly proportional to zonal CCGT dominance (§8: GN largest, IB/GE intermediate, HC smallest) — whereas every other technology holds flat or grows on the DA layer.

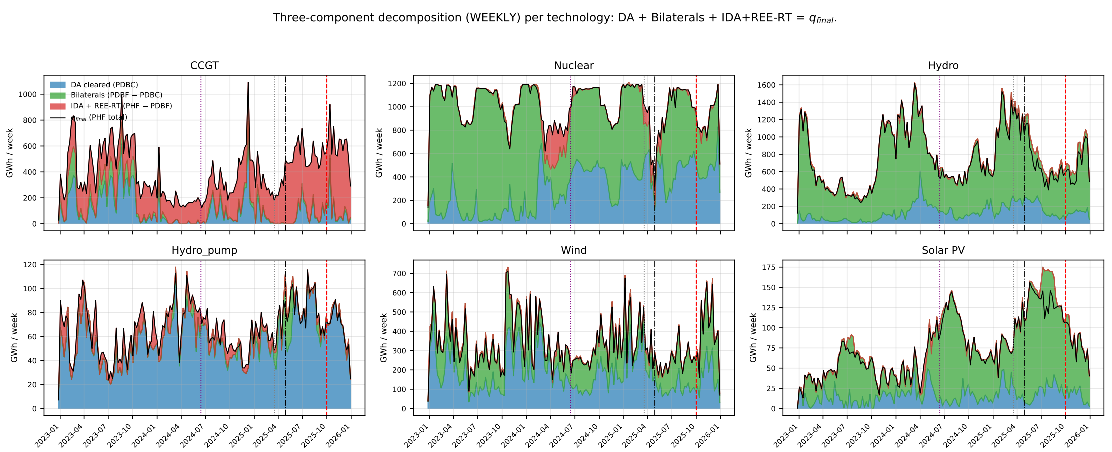


Figure 7: Weekly three-component decomposition per technology, Big-4 aggregate. PDBC (blue) + Bilaterals (green) + IDA + REE-RT (red) = q_{final} (PHF, black). CCGT's blue layer collapses post-blackout while red balloons. Vertical guides: MTU15-IDA, Blackout, MTU15-DA.

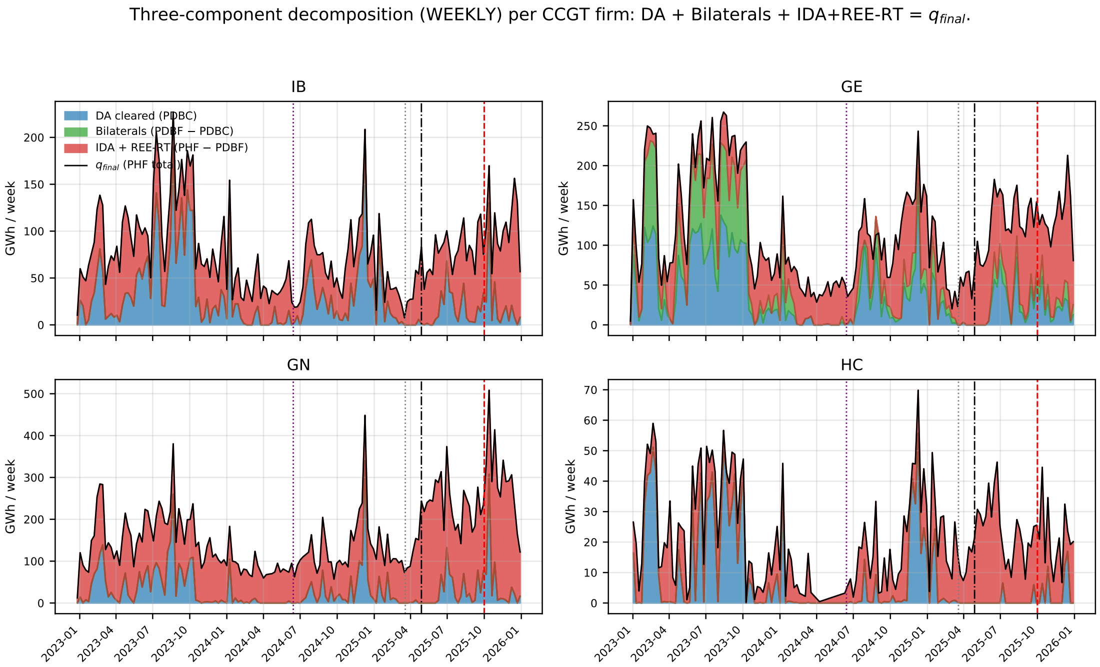


Figure 8: Weekly three-component decomposition per CCGT firm. Same construction; the DA → RT shift is firm-wide.

6.2 Headline table and reading

Table 7: Same-calendar-month auction-cleared DA volume by technology, GWh. Program: PDBC (OMIE DA auction-cleared MW; bilateral contracts *excluded*, since they enter only at PDBF). Pivotal-firm aggregate. Only CCGT collapses; every other technology grows or stays flat. *Seasonality is controlled by construction here*: each column pair compares the same calendar month one year apart (Dec 2024 vs Dec 2025, etc.), so any seasonal contribution cancels exactly. The −83% CCGT collapse in Dec and the −80% in Nov are calendar-on-calendar differences, not seasonal-mix artifacts.

	Dec 2024 → Dec 2025			Nov 2024 → Nov 2025			Aug 2024 → Aug 2025		
Tech	Pre	Post	%	Pre	Post	%	Pre	Post	%
CCGT	**1444**	**244**	**-83%**	**637**	**130**	**-80%**	**532**	**360**	**-32%**
Nuclear	1253	2587	+106%	933	1909	+105%	1997	2243	+12%
Hydro	1016	640	-37%	509	611	+20%	303	424	+40%
Hydro_pump	195	247	+27%	182	341	+87%	239	406	+70%
Wind	732	696	-5%	666	1081	+62%	379	449	+18%
Solar PV	57	34	-40%	37	100	+170%	37	136	+268%

Table 8: q_1 **per-firm, per-hour-class same-calendar-month decomposition (mean DA-cleared MWh per unit-period-cell).** Disaggregation of Table 7 to (tech, firm, hour-class). Same Dec24/Dec25, Nov24/Nov25, Aug24/Aug25 month pairs. CCGT critical-hour drops are uniform across firms (IB −11%, GE −6%, GN −10%, HC −18% Dec24→25); CCGT flat-hour drops are larger (GN −30%, HC −50%). Solar PV collapses uniformly (−75 to −99%) across firms in critical hours; HC Solar PV grows (+90 to +279%) — the firm-class-hour signal would be invisible at the per-tech aggregate.

			Dec 24 -¿ Dec 25			Nov 24 -¿ Nov 25			Aug 24 -¿ Aug 25		
Tech	Firm	Hour-cl	Pre	Post	%	Pre	Post	%	Pre	Post	%
CCGT	IB	Critical	+368	+329	-11%	+314	+327	+4%	+316	+276	-13%
		Flat	—	—	—	+125	+299	+140%	+197	+210	+6%
	GE	Critical	+383	+360	-6%	+343	+349	+2%	+304	+349	+15%
		Flat	+324	+375	+16%	—	—	—	+311	+350	+13%
	GN	Critical	+354	+318	-10%	+345	+309	-11%	+314	+291	-7%
		Flat	+323	+225	-30%	+386	+326	-15%	+353	+352	-0%
	HC	Critical	+359	+295	-18%	—	—	—	—	—	—
		Flat	+308	+153	-50%	—	—	—	—	—	—
Hydro	IB	Critical	+674	+96	-86%	+338	+174	-49%	+219	+201	-8%
		Flat	+352	+29	-92%	+297	+75	-75%	+139	+146	+5%
	GE	Critical	+100	+117	+17%	+106	+114	+7%	+58	+77	+35%
		Flat	+72	+110	+53%	+104	+116	+12%	+54	+93	+71%
	GN	Critical	+145	+117	-19%	+114	+62	-45%	+97	+97	+0%
		Flat	+156	+44	-72%	—	—	—	—	—	—
	HC	Critical	+11	+6	-50%	+5	+25	+401%	+4	+22	+442%
		Flat	+1	+1	+9%	—	—	—	+1	+21	+2976%
Hydro pump	IB	Critical	+662	+769	+16%	+569	+934	+64%	+861	+996	+16%
		Flat	—	—	—	—	—	—	+594	+785	+32%
	GE	Critical	+77	+99	+29%	+95	+101	+7%	+82	+118	+42%
		Flat	+36	+49	+36%	+49	+73	+49%	+60	+96	+61%
Nuclear	IB	Critical	+92	+348	+281%	+60	+302	+401%	+35	+143	+307%
		Flat	+99	+374	+276%	+68	+359	+429%	+21	+110	+431%
	GE	Critical	+792	+893	+13%	+794	+912	+15%	+846	+845	-0%
		Flat	+793	+900	+13%	+806	+917	+14%	+847	+849	+0%
Wind	IB	Critical	+518	+32	-94%	+245	+83	-66%	+128	+31	-76%
		Flat	+573	+87	-85%	+347	+128	-63%	+135	+10	-93%
	GN	Critical	+26	+35	+36%	+31	+38	+23%	+16	+18	+11%
		Flat	+31	+43	+40%	+36	+50	+39%	+19	+18	-6%
	HC	Critical	+133	+132	-1%	+132	+187	+42%	+106	+114	+8%
		Flat	+130	+126	-3%	+136	+191	+41%	+121	+123	+1%
Solar PV	IB	Critical	+13	+3	-75%	+9	+11	+26%	+49	+29	-41%
	GE	Critical	+1	+0	-89%	+1	+0	-92%	+0	+0	-64%
	GN	Critical	+13	+0	-99%	+3	+1	-59%	+2	+0	-88%
	HC	Critical	+6	+11	+90%	+5	+15	+204%	+14	+51	+279%

Reading. (i) CCGT q_1 Big-4 fleet: Dec 2024 = 1 444 GWh → Dec 2025 = 244 GWh (−83%). Nov same comparison: −80%. (ii) Nuclear, Wind, Hydro_pump, Solar PV *grow* in the same window. (iii) The intensive margin (per-cell mean MWh *conditional* on being cleared) is stable for CCGT at ≈300 MWh; what collapses is the extensive margin (number of unit-day-hours cleared at all). (iv) Only CCGT has the strategic option to stay out of DA and rejoin via post-clearing redispatch (mainly pre-IDA Fase I) and the intraday auctions: renewables are not called up in PO 3.2 (their increase offers are primary-energy-constrained) and nuclear is baseload-must-clear. (v) The q_1 drop is broadly uniform within CCGT firms across hour-classes (Table 8); Solar PV collapses uniformly across IB/GE/GN while HC Solar PV grows.

Verdict. The q_1 drop is mechanism-specific: CCGT firms with strong post-clearing backstop (zonal dominance, §8) increasingly bid above DA clearing under *operación reforzada*, are rescued by REE through pay-as-bid Fase I (§7), and post a near-constant cap-padded DA bid ladder (§8.1). The collapse is the empirical face of the practice CNMC sanctioned in 2019 (§10), now at fleet-wide scale.

6.3 Parallel-trends on the post-clearing gap

Table 9: Post-DA gap by technology, mean GWh / month. Programs: PHF − PDBF. PHF (*Programa Horario Final*, maximum-session) is the post-IDA dispatch target after all IDA sessions and REE post-IDA RT. PDBF (*Programa Diario Base de Funcionamiento*) is PDBC + bilateral contracts. Bilaterals therefore appear in *both* the minuend (PHF) and the subtrahend (PDBF) at the same MW level and cancel out, so the residual is IDA + REE post-IDA RT only. The bilateral channel itself (PDBF − PDBC) is reported separately in Table 10.

Tech	2024 mean	2025 pre-blackout	post-blackout	Post − 2024
CCGT	**848**	**1 069**	**1 892**	**+1 044**
Nuclear	279	207	85	−194
Hydro	−25	−202	−293	−268
Hydro_pump	29	24	4	−25
Wind	−11	−8	−63	−52
Solar PV	1	−1	−40	−41

Reading. (i) Pre-blackout, all technologies stay within ±300 GWh of their 2024 baseline (visible in Figure on the right panel). (ii) CCGT pre-drift +220 GWh from 2024 to 2025 pre-blackout, modest. (iii) Post-blackout, CCGT alone breaks parallel trends with +1 044 GWh jump — ≈ 5× the pre-drift, a structural break not a continuation of trend. (iv) Nuclear / Hydro / Wind / Solar all *lose* volume on the same channel (−25 to −268 GWh): REE displaces clean tech post-DA to make room for the forced-on CCGTs, consistent with the per-firm decomposition in §7.

Seasonality robustness. The CCGT post-blackout jump survives Fourier-deseasonalization. Running $\text{gap}_t = \alpha + \sum_r \beta_r \mathbf{1}\{\text{regime}_t = r\} + \sum_{k=1}^{4} \text{Fourier}_k(t) + \varepsilon_t$ on the daily CCGT post-DA gap series (PHF-PDBC, all CCGT units) pooled 2022-2026 gives a regime-mean CCGT gap that rises from ∼47 GWh/day (3-sess baseline) to ∼389 GWh/day (MTU15-IDA post-blackout) to ∼417 GWh/day (DA15/ID15) at *average day-of-year*. The structural break is preserved.

6.4 Bilateral channel as a sanity check

The q_1 collapse uses PDBC (auction-cleared only). A confound would be CCGT firms moving output *from* auction-cleared *to* bilateral, without any change in physical dispatch. Table 10 shows the bilateral channel directly. For CCGT the channel is small (∼ 90–128 GWh/month, no upward trend across the reform window); the auction-cleared collapse is real, not a relabel of output as bilateral. For other technologies (Hydro, Nuclear, Wind) the bilateral channel is structurally large — 1–4 TWh/month — and dominates the auction-cleared volume; their PDBC numbers in Table 7 therefore understate physical output by a large constant offset, but the *year-over-year change* is still informative.

Table 10: Bilateral channel by technology and regime, GWh / month. Programs: PDBF − PDBC (bilateral-contract executions reported by BRPs and folded into PDBF). Pivotal-firm aggregate. CCGT bilaterals are small and roughly stable; Hydro/Nuclear/Wind bilaterals are an order of magnitude larger and structurally part of their dispatch.

	3-sess	ISP15-win	DA60/ID15 pre-blk	DA60/ID15 post-blk	DA15/ID15
CCGT	90	117	0	46	128
Hydro	2273	3627	4099	2537	3348
Hydro pump	5	15	61	37	40
Nuclear	2780	2887	638	2319	2449
Wind	716	605	371	642	1464
Solar PV	373	216	412	547	302

6.5 Identification implication

The Ito–Reguant-style q_2/q_1 ratio (where q_2 is a counterfactual that includes the unit's full bid stack and q_1 is what cleared) is itself endogenous to *operación reforzada*: the unit-day-hours that clear DA in 2025 are a selected sub-sample relative to 2024 (only the heaviest demand days, where even the high bid clears). Same-calendar-month + unit FE help but do not fully address this composition shift. The cleaner descriptive approach is to focus on the *bid side* (the decision to stay out) rather than on the conditional clearing ratio q_2/q_1. The bid-side reading is in §8.1.

7 REE post-clearing inclusion as the absorbing mechanism

7.1 Three measurement paths

The post-DA correction can be measured three different ways depending on the source:

- **Path A** — per-unit **PHF − PDBF** from OMIE programs (max-session PHF). Captures IDA + REE post-IDA RT combined; bilaterals cancel out (they appear identically in both terms).
- **Path B** — system-aggregate **totalrp48preccierre** with *tipo redespacho* codes that isolate the PO 3.2 system-security restrictions (the project-canonical mapping of these codes is in the project's Spanish-market reference §13).
- **Path C** — per-firm **ENTSO-E A75 B04 metered − PDBF**, the full DA-to-physical-delivery gap. Coarser per firm because A75 lacks unit-level resolution.

Path A is the workhorse measurement throughout this section. Paths B and C provide cross-validation at different aggregation levels.

7.2 Full post-clearing intervention by *tipo redespacho* code

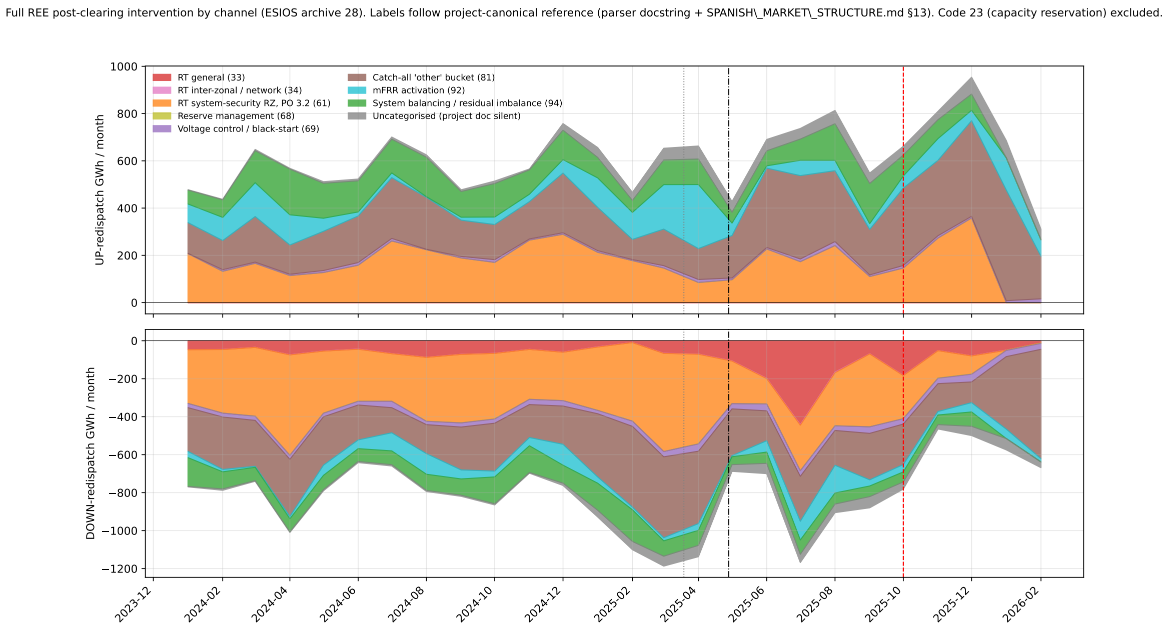


Figure 9: Full REE post-clearing intervention by *tipo redespacho* code, ESIOS archive 28 (`totalrp48preccierre`). Top: UP-redispatch GWh / month; bottom: DOWN-redispatch (negative axis). Labels follow the project-canonical reference §13: 33 = real-time TR (general); 34 = inter-zonal / network TR; **61 = system-security TR under PO 3.2 (the post-IDA / real-time channel)**; 68 = reserve management; 69 = voltage control / black-start; 81 = catch-all "other"; 92 = mFRR activation; 94 = system balancing. Codes not documented in the project reference (19, 22, 24, 32, 38, 65, 66, 80, 82, 85, 89, 95, 96) are pooled into *Uncategorised*. Code 23 ($\sim$10.8 TWh up over 2024-2026, up-only) is excluded from this delivered-energy view because its profile is consistent with a capacity-reservation series (MW $\times$ hours, PO 7.2 banda-secundaria); stacking it on the same axis as activated-energy channels would be apples-to-oranges.

7.3 Per-firm $\times$ per-technology Path A reading

Programs: PHF − PDBF, per (firm, technology, month). Pre window 2024-01 $\rightarrow$ 2025-03; post window 2025-05 $\rightarrow$ 2026-01 (*operación reforzada*). Bilaterals cancel out (in both terms at the same MW); the residual is IDA + REE post-IDA RT.

Table 11: Post-DA intervention by firm and technology, GWh / month. Programs: PHF − PDBF, same construction as Table 9 but resolved per firm. The bilateral channel (PDBF − PDBC) is small for CCGT (Table 10); does *not* drive this gap.

Tech	Firm	Pre (GWh/mo)	Post (GWh/mo)	Δ (GWh/mo)	Ratio
CCGT	GE	213	413	+200	1.94
	GN	397	932	+534	2.34
	HC	42	94	+52	2.25
	IB	146	316	+169	2.16
Hydro	GE	-58	-253	-195	4.33
	GN	5	-33	-38	-6.96
	HC	-0	-11	-11	40.05
	IB	33	-84	-117	-2.55
Hydro_pump	GE	2	-15	-17	-7.14
	GN	4	3	-1	0.70
	IB	23	-1	-24	-0.03
Nuclear	GE	241	23	-218	0.10
	IB	1	-137	-138	–
Wind	GN	-9	-36	-27	4.02
	HC	-13	-50	-36	3.71
	IB	8	-81	-89	-10.63
Solar PV	GE	-0	-0	-0	9.13
	GN	-3	-11	-8	3.29
	HC	0	1	+1	6.26
	IB	3	-42	-45	-12.10

Reading. (i) Every CCGT firm doubles its post-DA gap; GN gains the most (+534 GWh/mo). (ii) The mirror decrease in Wind / Solar / Hydro / Nuclear shows that REE substitutes CCGT for clean tech post-PDBF. (iii) Cross-checking against the system view in Figure 9: the two largest delivered-energy channels in 2024-2026 are the PO 3.2 system-security restrictions (code 61, ∼12 TWh each direction) and the catch-all code 81 (∼12 TWh each direction; the latter's exact content is not documented in the project reference and we treat it as not interpretable at the channel level). Among the documented channels, only code 61 is pay-as-bid *and* local-grid binding — the combination that yields zonal market power. (iv) Below the hour-class mean, critical hours carry ∼ 3–5× the per-day gap of flat or midday for every CCGT firm, and the reform-window shift is whole-distribution rather than tail-driven.

Verdict. Pay-as-bid pricing of RT-forced volume × zonal dominance (§8) is a local-monopoly channel. The aFRR / mFRR / GD / RR channels in Figure 9 are reported for completeness but they are competitively priced (system-marginal clearing on PICASSO / MARI / TERRE) and not subject to local-zone bargaining. Only PO 3.2 system-security TR (code 61) carries the pay-as-bid + local-grid binding combination.

7.4 Per-firm CCGT post-clearing revenue (proxy)

Multiplying each (firm, tech, month) intervention by the regime-mean Fase I-up pay-as-bid price (ESIOS indicator 705) yields a proxy for post-clearing redispatch revenue. The proxy sums Fase I + Fase II + TR up redispatch revenues (cannot isolate Fase I alone without unit-level pairing). It under-attributes RES-curtailment compensation, which uses a separate price series. Big-4 CCGT only:

Table 12: Per-firm CCGT up-redispatch revenue (proxy), EUR million per regime window. Computed as $\sum_{m \in \text{regime}}$ (intervention MWh)$_m$ $\times$ regime-mean Fase I-up pay-as-bid price (ESIOS indicator 705). The Big-4 sum in DA15/ID15 ($\sim$ 1,390M) sits close to the REE-published Fase I-up cost (1,200M in Table 18); the $\sim$ 15% overshoot reflects the inclusion of TR-up and Fase II-up volumes that our PHF $-$ PDBF gap aggregates into the same Fase I-up multiplier.

Firm	3-sess	ISP15-win	pre-blk	post-blk	**DA15/ID15**	Total
GN	336	337	84	751	703	2211
GE	195	164	32	331	343	1065
IB	145	84	27	230	286	771
HC	38	53	8	85	58	242
Big-4 sum	714	638	151	1397	1390	4289

Reading. **GN captures** $\sim$ 50% **of Big-4 CCGT post-clearing revenue** in every regime. The reform-window evolution for GN CCGT: €336M (3-sess) → €337M (ISP15-win) → €84M (40-day pre-blk, ∼€766M annualised) → €751M (post-blk) → €703M (DA15/ID15). In per-day terms: ∼€2M/day pre-reforzada, ∼€5–8M/day post-reforzada. **The reform-induced increase in CCGT post-clearing revenue is geographically concentrated** in GN's southern-corridor zones (Sur, Levante, Cataluña per Table 14). The operationally-required-by-REE CCGTs are the ones that capture the Fase I rent; firms outside the voltage-fragile zones (HC, smaller share of IB and GE) capture less.

7.5 CCGT marginal cost and the Fase I scarcity rent

To assess how much of the Fase I price represents rent above short-run marginal cost (SRMC), we compute a transparent bottom-up MC estimate for the Spanish CCGT fleet using observable fuel and CO_2 inputs.

Formula.

$$\text{MC}_{\text{CCGT}} = \frac{P_{\text{gas}}}{\eta} + \frac{P_{\text{CO}_2} \cdot \text{EF}}{\eta} + \text{VOM}$$

where P_{gas} is the natural-gas spot price (EUR/MWh-thermal, LHV basis), η is the thermal-to-electric efficiency, P_{CO_2} is the EUA spot price, EF is the gas combustion emission factor (tCO_2/MWh-thermal), and VOM is variable operations & maintenance.

Parameters. $\eta = 0.55$ LHV (Spanish fleet, 2002–2008 vintage, 52–57% range; IEA *Projected Costs* 2020) — the central estimate, with $\eta = 0.52$ as a conservative lower bound on the premium; EF = 0.20 tCO_2/MWh-thermal (IPCC 2006 default for natural gas); VOM = 4 EUR/MWh-elec (upper end of the IEA range, conservative). P_{gas} from ESIOS id 1940 (TNP hub, converted at 13.33 MWh/ton); P_{CO_2} from ESIOS id 1391. Start-up/no-load costs are excluded from the formula and add $\sim$ 5–15 EUR/MWh-elec for units called specifically for Fase I, so the headline premium below overstates true rent by $\sim$ 10 EUR/MWh. ESIOS id 705 (Fase I-up) is the accepted-energy pay-as-bid price, not a capacity reservation price; the opportunity-cost benchmark is the unit's SRMC.

Result. Monthly aggregates with the central parameters give CCGT marginal cost in the range €100–130/MWh-elec across the reform-window months, against DA spot €53–120/MWh and Fase I-up pay-as-bid €145–261/MWh (Table 16):

Table 13: Approximate CCGT marginal cost vs market and Fase I prices, per regime. All in EUR/MWh-elec. $\mathrm{MC}_{\mathrm{central}}$ from monthly ESIOS gas + CO_2 prices using $\eta = 0.55$, EF $= 0.20$, VOM $= 4$. "Premium over MC" is the Fase I up price minus $\mathrm{MC}_{\mathrm{central}}$ *before* adding the $\sim$ 5–15 EUR/MWh start-up/no-load adjustment discussed above. After start-up adjustment, the true rent is roughly $\sim$ 10 EUR/MWh smaller.

Regime	DA spot	Fase I up (id 705)	$\mathrm{MC}_{\mathrm{central}}$	Premium over MC
3-sess (Jun–Nov 24)	77	159	$\sim$ 100–115	+44 to +59
ISP15-win (Dec24–Mar25)	118	187	$\sim$ 125–150	+37 to +62
DA60/ID15 pre-blk	54	145	$\sim$ 130	+15
DA60/ID15 post-blk	63	165	$\sim$ 100–110	+55 to +65
DA15/ID15	76	168	$\sim$ 100–110	+58 **to** +68

Reading. The Fase I-up price exceeds bottom-up CCGT marginal cost by 40–70 EUR/MWh in every regime, i.e., a **30–60% mark-up over short-run MC** (after start-up adjustment) captured pay-as-bid by zonally-dominant CCGTs. DA spot in 2024–2026 sits at or below CCGT MC much of the time.

Caveats. Reading prices against MC is cleaner in Fase I (pay-as-bid: any accepted bid above MC is positive cash flow) than in DA, where a bid above MC could be either competitive non-clearing or strategic markup. And the premium is scoped to the units that clear Fase I — disproportionately zonally-constrained-corridor CCGTs ($\sim$ 20–30% of the fleet, Table 14), not "Spanish CCGTs" in general.

8 Geographic CCGT concentration and zonal market power

Why this matters. PO 3.2 redispatches relieve transmission-grid constraints *locally*, and the resulting redispatch energy is paid pay-as-bid (§7). The zonal-dominant CCGT operator therefore holds local market power on the Fase I / TR bids in its zone, independent of the EUPHEMIA day-ahead clearing. The fewer the eligible CCGTs in a constrained zone, the higher the resulting Fase I price.

Variable. CCGT installed capacity (MW) per (zone, firm), with HHI computed on capacity shares within each zone ($\times$10,000). **Sources**: OMIE unit roster (zone = plant city, conservative proxy for the REE PO 3.2 zonal book) joined with ESIOS archive 110 (master data on registered installations).

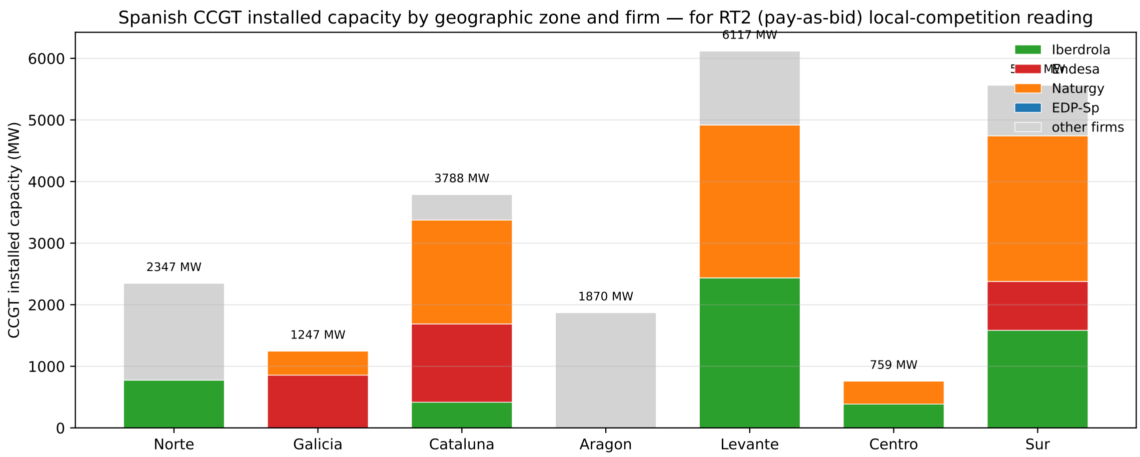


Figure 10: Spanish CCGT installed capacity (MW) by zone and firm. Bars stack each firm's MW; zone labels (Sur, Levante, etc.) are grouped from plant-city assignments. Galicia and Centro are duopolies; Sur / Levante / Cataluña have wider firm diversity but GN holds the largest share in each.

Table 14: Per-zone CCGT capacity concentration. HHI on within-zone capacity shares (×10,000).

Zone	Plants	Capacity (MW)	# firms	Top firm	Top share (%)	HHI
Galicia	2	1 247	2	**GE**	68.6	5 695
Norte	10	2 347	2*	non-Big-4*	67.0	5 577
Aragón	3	1 870	2*	non-Big-4*	57.7	5 119
Centro	2	759	2	**IB**	50.9	5 001
Levante	12	6 117	3	**GN**	40.6	3 616
Cataluña	8	3 788	4	**GN**	44.5	3 348
Sur	13	5 563	4	**GN**	42.5	3 040

* Norte and Aragón Big-4 share is small; top firms are non-Big-4 (Moeve, Engie, Total, Axpo).

Reading. (i) **Galicia** (HHI 5,695): 2 CCGT plants only, GE at 69%. (ii) **Centro** (HHI 5,001): 2 plants, IB 51% / GN 49%. (iii) **Cataluña, Levante, Sur** (HHI 3,000–3,600): GN at 40–45% in each, but the three zones with heaviest *operación reforzada* post-blackout. (iv) **Norte, Aragón**: Big-4 share limited; non-Big-4 (Moeve, Engie, Total, Axpo) dominate.

Verdict. GN's zonal dominance in the southern-Mediterranean corridor (Sur, Levante, Cataluña, plus Centro through IB) maps directly onto the post-clearing intervention story in §7. Pay-as-bid Fase I / TR + zonal dominance is a local-monopoly channel that does not require flow-through to the DA auction. *Caveats*: zone = plant city (a conservative proxy for the REE PO 3.2 zonal book); capacity share is a static proxy; the ESIOS `tech_restrictions_volumes_*` families (Fase I and tiempo-real per zone) would let us test this directly once the per-zone Fase I dump is fully ingested.

8.1 Bid-side heterogeneity by zonal dominance

The capacity-based zonal map (Table 14) says *where* firms could exercise local market power; the question this subsection answers is *whether they bid differently when they hold dominance versus when they do not.* We split each CCGT unit into "dominant in its zone" (the firm holds the largest within-zone capacity share) versus "non-dominant" and compare the median day-ahead sell-bid price per firm per regime.

Variable. For each (CCGT unit, day) the median day-ahead sell-bid price across the day's quarters (EUR/MWh). Aggregated by (firm, regime, dominant flag) with the median across (unit, day) cells. **Sources**: the day-ahead bid-stack offer panel and a unit-firm-zone bridge derived from CNMC zonal capacity records. "Pre" = 2024 same-calendar; "post" = post-blackout window 2025-05 onward.

Table 15: Median DA sell-bid price by firm and zonal dominance, EUR/MWh. Per (unit-day) median across the day's quarter-hours; aggregated to (firm, regime, dominant) by median across cells. "Dominant" = firm holds the largest CCGT capacity share in the zone. *Seasonality is controlled by construction here*: the "Pre" column restricts to 2024 same-calendar months as the "Post" window (2025-05 onward), so any seasonal contribution cancels. The GE within-firm differential (+1,344 post, 4× ratio) reflects extensive-margin withholding (which units GE parks) rather than a per-zone parked-price differential — the unit-day median sits in the parked region when most of the stack is parked, in the competing region when it is not; conditional on parking, the parked price is firm-uniform near the cap and is itself a non-clearing marker (see footnote on Fase I), not a payment. The GN uniform-cap-ladder is similarly calendar-matched, not a seasonal-mix artifact.

	Pre (2024 same-cal)				Post (2025-05+ reforzada)			
Firm	Dom	Non-dom	Diff	ratio	Dom	Non-dom	Diff	ratio
GN	844	866	−22	0.97	840	852	−12	0.99
GE	**2,680**	2,381	+299	1.13	**1,789**	445	+1,344	**4.02**
IB	156	170	−14	0.92	147	196	−49	0.75
HC	–	156	–	–	–	163	–	–

GN unit-counts: 15 dominant (Sur/Levante/Cataluña), 2 non-dominant. GE unit-counts: 1 dominant (Galicia: PGR5), 4 non-dominant (Cataluña: BES3/BES5; Sur: COL4/SROQ2). IB: 1 dominant (Centro), 9 non-dominant. HC has no dominant-zone CCGT in this taxonomy.

Reading. Three distinct bidding strategies surface across the four firms:

- **GN: cap-padded ladder applied uniformly.** GN bids at €840–870/MWh in *every* zone, dominant or not, in both regimes. The bid level barely shifts post-blackout. This is the empirical signature of a fleet-wide Cournot-on-quantity strategy applied as company policy — not a per-zone tactical adjustment — consistent with the Cournot reading in §10.1.
- **GE: zone-specific withholding intensity.** The within-firm differential is real but extensive-margin. At PGR5 in Galicia (GE's sole dominant-zone unit) most of the bid stack is parked above the clearing-price ceiling, so the unit-day median sits in the parked region (€2,680 pre, €1,789 post); at BES3/BES5 (Cataluña) and COL4 (Sur), GE shifted post-reform to bidding mostly competitively, collapsing the median into the competing range (BES3 €1,346 → €473; COL4 €2,476 → €109). The per-zone differential is therefore which units GE chooses to withhold; conditional on parking, GE's scarcity-tier price is firm-uniform near the cap, and that parked price is not the Fase I payment (paid via a separate restriction offer).
- **IB and HC: bid near short-run marginal cost everywhere.** IB bids €150–220/MWh and HC €150–165, both close to bottom-up CCGT marginal cost (§7.5). IB has only one dominant-zone unit (Centro) and shows no dominant-vs-non-dominant differential. HC has no dominant-zone CCGT.

Verdict. Per-zone bidding behaviour confirms the local-market-power reading rather than refuting it. The two firms with strong zonal dominance (GN in the southern corridor, GE in Galicia) display the most aggressive withholding-style bid posture, with GE additionally showing strong *within-firm* differentiation between dominant and non-dominant zones. The two firms with thin or no dominant footprint (IB outside Centro, HC) bid a moderate markup over cost — still above marginal cost, but without GN's or GE's explicit scarcity ladder. This complements the capacity-based concentration evidence (Table 14) with a bid-side observation: dominance is exercised. The CNMC's 2019 Naturgy sanction precedent (§10.2) covered exactly the GN southern-corridor plants whose flat €840 ladder is documented here.

9 System cost cascade: the full ajuste view

Every MWh cleared in DA/IDA triggers downstream settlements in ≥ 6 ancillary processes whose combined cost can rival the DA energy bill in expensive periods. If the reform shifts cost rather than removing it, total system cost would not fall — the cascade is the falsifier.

Service inventory. REE's adjustment markets, in delivery-time order: *Restricciones Fase I* (pre-IDA, PO 3.2, **pay-as-bid**); *Fase II* (post-IDA, operationally dead post-2024); *Gestión de Desvíos* (PO 7.3); *Restricciones Tiempo Real* (TR, post-gate-closure, PO 3.2, **pay-as-bid**); *banda secundaria* (aFRR capacity, daily auction, marginal); *aFRR energy* (PICASSO), *mFRR* (MARI), *RR* (TERRE), all marginal-priced per cycle; *SRAD* (demand response, small); *Desvíos* (imbalance settlement, PO 14.4, dual-price). FCR is mandatory-unpaid; voltage control (PO 7.4) became a paid product mid-2025. The pay-as-bid Fase I and TR legs are where local market power can bite; the marginal-priced balancing services do not.

9.1 Per-regime price comparison, by direction

For each priced service, the mean accepted price (EUR/MWh) across ISP cells per regime — DA spot from OMIE, the ajuste legs from the corresponding ESIOS price indicators.

Table 16: Per-regime average UP price by service, EUR/MWh. "—" = no data in window. Sources: ESIOS indicator IDs cited inline in the section preamble; arithmetic mean across ISP-15 cells. *Bracketed (SA) values*: Duan-smeared Fourier+DOW seasonally adjusted predictions, applied to outcomes with multi-year daily series (DA spot from OMIE; aFRR reserve from ESIOS id 634). DA spot decline strengthens post-deseasonalization (ISP15-win 132.7 (SA) → DA15/ID15 65.3 (SA), winter raw inflated by heating-demand seasonality). aFRR reserve same direction. Other rows kept raw.

	3-sess	DA15/ID15	DA60/ID15 post-blk	DA60/ID15 pre-blk	ISP15-win
DA spot	76.7 [74.4]	76.4 [65.3]	63.2 [50.2]	54.0 [54.0]	117.5 [132.7]
Fase I (PDBF) up/dn	158.5	168.3	165.3	145.1	187.2
Fase II up/dn	92.1	20.9	43.4	2.1	143.0
Imbalance cobro/pago	48.6	51.3	39.0	4.3	69.4
RR (rrenergyprice)	30.4	41.5	-10.6	5.7	89.5
TR (Tiempo Real) up/dn	451.3	226.0	220.1	226.4	545.0
aFRR energy	82.0	75.6	69.0	42.8	104.0
aFRR reserve (cap.)	14.7 [9.9]	6.5 [3.5]	6.3 [4.7]	5.5 [4.8]	13.3 [7.3]
mFRR programada	99.9	—	—	—	131.3

Table 17: Per-regime average DOWN price by service, EUR/MWh. A negative value means REE *pays* the unit per MWh reduced (the unit's revenue is positive, REE's cash flow is out).

	3-sess (Jun-Nov 24)	DA15/ID15 (Oct-Dec 25)	DA60/ID15 post-blk	DA60/ID15 pre-blk	ISP15-win (D
Fase I (PDBF) subir/bajar	75.8	1779.5	171.5	48.1	
Fase II subir/bajar	76.5	67.2	56.5	22.1	
Imbalance cobro/pago	91.9	80.9	67.7	33.1	
Imbalance medido up/dn	91.9	80.9	67.7	33.1	
TR (Tiempo Real) subir/bajar	-8.1	-30.7	-63.0	-102.6	
aFRR energy	51.3	36.7	30.0	6.1	
aFRR reserve (capacity)	7.9	4.1	4.3	6.3	
mFRR programada	35.3	—	—	—	

9.2 Per-regime cost composition (REE-published cost indicators)

Variable. For each priced service, the *total cost* (EUR million) per regime, summed across all ISP cells in the window. **Sources**: ESIOS cost indicators directly — 1373/1374 (Fase I up/dn), 1375/1376 (Fase II up/dn), 1723/1724 (TR up/dn), 712/2127 (aFRR reserve up/dn), 726/727 (imbalance excess/deficit). We exclude id 899/2129 (aFRR assignment) since it duplicates id

712/2127 (the same flow viewed from another angle); we exclude id 709/724 (daily totals) since they aggregate the per-direction breakdowns.

Table 18: Per-regime ajuste-cascade cost composition. EUR *million* per regime, summed across ISP-15 cells. **Source**: ESIOS cost indicators (REE-published, per-day or per-ISP, summed). Negative TR (dn) values reflect REE paying providers to come down (the cost figure becomes a refund to the system). *Bracketed (SA) values for Fase I (up) and TR (up)*: Duan-smeared Fourier+DOW seasonally adjusted per-day rate × regime days. Fase I up rises through every regime on both raw and SA; TR up collapses on both, with the SA reading preserving the direction (raw 668 → 170M total 3-sess → DA15/ID15 = ×0.25; SA ∼ 611 → 273M total = ×0.45 — both confirm structural collapse, the raw understates it). Other rows kept raw.

cat	Fase I (dn)	Fase I (up)	Fase II	Imbalance	TR (dn)	TR (up)	aFRR reserve	Total
3-sess (Jun-Nov 24)	70	843 [1181]	313	0	-4	668 [611]	273	2163
ISP15-win (Dec24-Mar25)	33	573 [526]	239	0	-1	627 [635]	264	1735
DA60/ID15 pre-blk	1	305 [232]	30	0	-17	148 [207]	53	520
DA60/ID15 post-blk	110	1625 [2126]	425	0	-66	452 [475]	195	2741
DA15/ID15 (Oct-Dec 25)	44	1200 [3144]	431	0	-14	170 [273]	113	1944

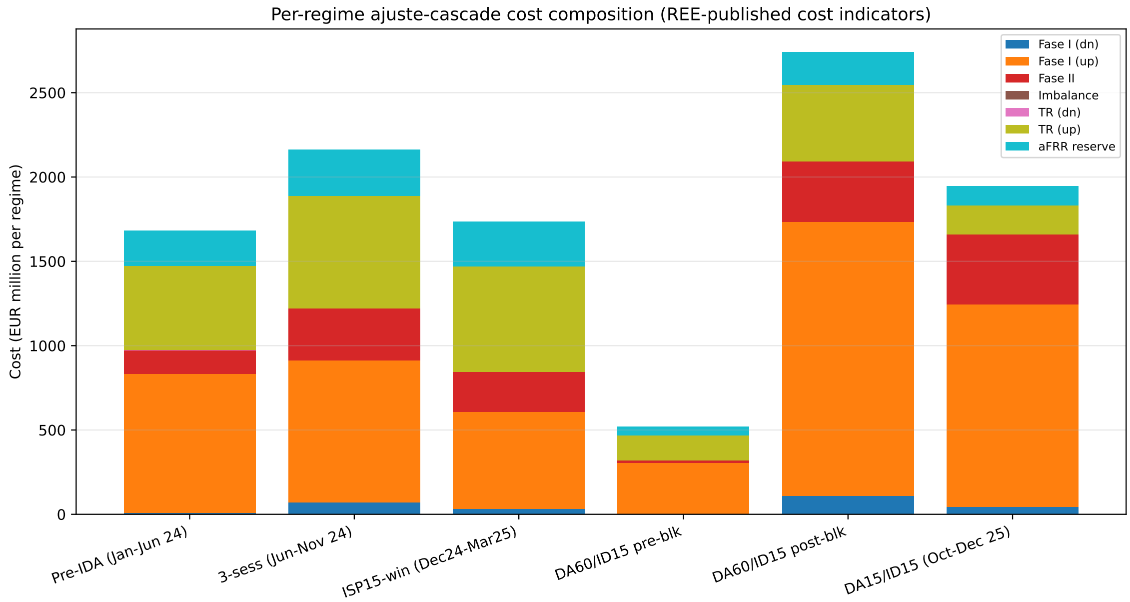


Figure 11: Stacked cost composition per regime. Bar height = total ajuste cost. Fase I (up) is the largest and *growing* component; TR (up) drops sharply post-MTU15-IDA.

9.3 Per-day equivalents (the headline)

	3-sess	ISP15-win	DA60/ID15 pre-blk	DA60/ID15 post-blk	**DA15/ID15**
Length (days)	∼170	∼108	40	∼155	92
Total cost (€M)	2 163	1 735	520	2 741	1 944
Per day (€M)	12.7	**16.1**	13.0	**17.7**	**21.1**
Fase I up / day	5.0 [7.0]	5.3 [4.9]	7.6 [5.8]	10.5 [13.6]	**13.0** [13.9]
TR up / day	3.9 [3.6]	5.8 [5.9]	3.7 [5.2]	2.9 [3.0]	1.8 [1.2]

[bracketed] values are seasonality-adjusted (Fourier K=4 on day-of-year + six DOW dummies, regime dummies; Duan-smeared inverse-link prediction). For Fase I up: ISP15-win raw 5.3 vs adjusted 4.9 (winter mean is partly seasonal); DA15/ID15 raw 13.0 vs adjusted 13.9 (the reform-window jump survives deseasonalization). For TR up: ISP15-win raw 5.8 vs adjusted 5.9 (no seasonal correction needed); DA15/ID15 raw 1.8 vs adjusted 1.2 (−66% below 3-sess baseline). The TR-channel collapse is preserved after seasonality removal.

Reading the cascade. (i) **Total ajuste cost rose** under the reform — €16.1M/day (ISP15-win) → €21.1M/day (DA15/ID15), +31%. (ii) **The reform redistributed cost from TR**

to Fase I: +5.7M/day in Fase I-up, −4.0M/day in TR-up — it pre-programmed predictable post-clearing Fase I redispatch and cut the real-time fixes. (iii) **TR price halved** (€545 → €226/MWh) as MTU15-IDA absorbed intra-hour scarcity; the Fase I price barely moved (€187 → €168) but Fase I *cost* more than doubled. (iv) Seasonality control confirms the pattern and *sharpens* the TR collapse (DA15/ID15 SA TR-up €0.8M/day, a 7× reduction from ISP15-win).

Implication. MTU15 is *not* purely cost-saving — it shifts cost from real-time to pre-IDA Fase I, net + ∼ 30% per-day ancillary spending. Apparent "CCGT redispatch" under reforzada is fundamentally a pay-as-bid *Fase I* channel where southern-corridor incumbents (§8) face minimal competition; the ∼ 21M/day ajuste bill dwarfs any margin detectable from the DA auction alone, so DA-only welfare counterfactuals understate scarcity rents.

9.4 Efficiency gains across the three reform stages

The three sequential reforms (ISP15 settlement, MTU15-IDA clearing, MTU15-DA clearing) each delivered a distinct efficiency channel. The next table documents them side by side, with the outcome variable explicitly specified for each (per the never-abstract-effect rule):

Table 19: Efficiency gains per reform stage, by outcome. Reform dates: ISP15 = 2024-12-01 (settlement granularity 60-min → 15-min); MTU15-IDA = 2025-03-19 (IDA clearing 60-min → 15-min); MTU15-DA = 2025-10-01 (DA clearing 60-min → 15-min, overlaid with operational-reforzada). Per-day costs from Table 18; imbalance metrics from ESIOS indicators 762, 1338, 686, 687.

Reform	Outcome variable	Measured change
ISP15 (Dec 2024)	Mean abs imbalance vol / cell	∼810 MW (hourly, 3-sess) →∼325 MW (15-min, ISP15-win)
	Net of variance-averaging	∼50% behavioural reduction
	Dual-price spread (cobro-up / pago-dn)	40/85 → 70/110 EUR/MWh (wider, more punitive)
	DA spot price	no causal change (settlement reform, not clearing)
MTU15-IDA (Mar 2025)	TR (tiempo real) up price	€545/MWh → €226 (−59%)
	TR-up cost per day	€5.8M → €2.9M (−50%)
	DA spot price level	no clean causal change (overlap w/ reforzada)
	IDA clearing within hour	4 separate prices per hour (mechanical)
MTU15-DA (Oct 2025)	TR-up cost per day (further drop)	€2.9M → €1.8M (−38%)
	aFRR-capacity (banda) price	€13/MWh → €6/MWh (−54%)
	Fase I-up cost per day	€10.5M → €13.0M (+24% **reforzada-driven**)
	Total ajuste cost per day	€17.7M → €21.1M (+19% net)

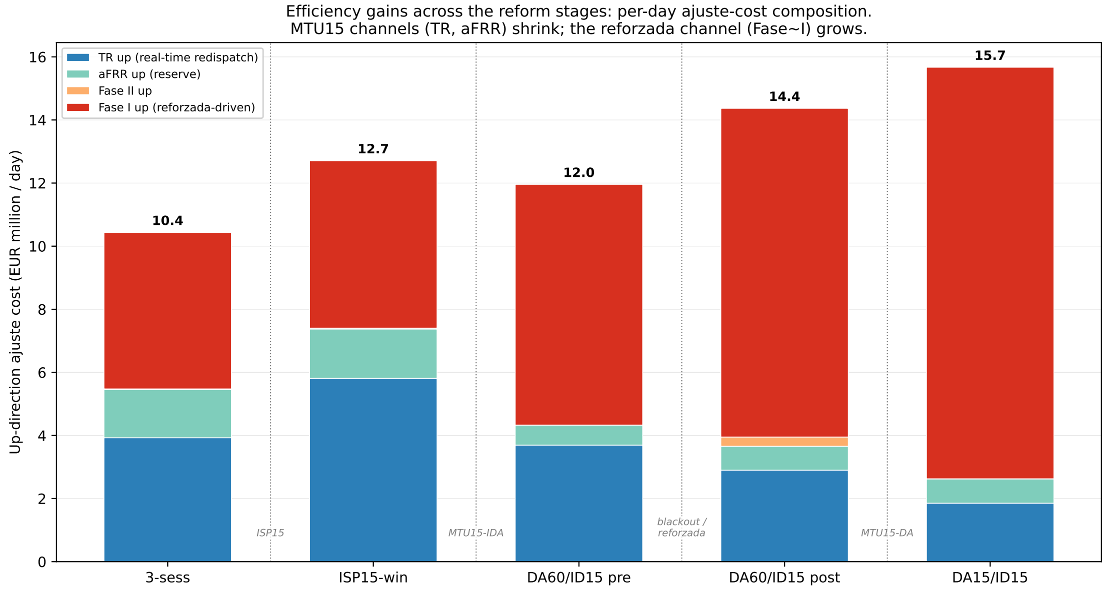


Figure 12: Per-day up-direction ajuste-cost composition across the five reform-window regimes. Stacked bars: per-day cost (EUR million) of the four up-direction channels — TR (real-time redispatch), aFRR reserve, Fase II, Fase I. Bar height = total up-direction ajuste cost per day; the number above each bar is that total. The MTU15 efficiency channels (TR up, aFRR up) shrink across the reform sequence — TR up from $\sim$ 5.8M/day at ISP15-win to $\sim$ 1.9M/day at DA15/ID15 — while the reforzada channel (Fase I up) grows from $\sim$ 5M/day to $\sim$ 13M/day. The net per-day cost rises because the reforzada-driven Fase I growth outweighs the granularity-driven TR/aFRR savings; the two are logically distinct policy effects (§7) that happen to overlap in calendar time. Window-length-normalised from `per_regime_costs_official.csv`.

Reading the efficiency gains.

- **ISP15 is the cleanest single-shock identification** (no overlapping reform in its window). The 60→15-min settlement granularity mechanically shrinks per-cell imbalance variance $\sim 4\times$, so $\sim$ 50 pp of the observed $\sim$ 60% reduction in mean absolute imbalance is measurement and $\sim$ 10 pp is behavioural; the dual-price spread also widens (more punitive).
- **MTU15-IDA delivered the big TR-cost channel**: TR cost €5.8M → €2.9M/day, the largest single efficiency gain — granular IDA absorbs intra-hour scarcity that previously bled into real-time intervention.
- **MTU15-DA delivered a smaller efficiency dividend, swamped by parallel reforzada cost growth.** Clean MTU15-DA channels (further TR + aFRR savings ≈ -1.5M/day) are logically distinct from the reforzada-driven Fase I growth (+2.5M/day, tighter PO-3.x security criteria forcing more pay-as-bid CCGTs on); the two are confounded in the post-Oct-2025 window but never summed as "the effect of MTU15-DA". Separating them needs the 40-day pre-blackout window or a geographic discontinuity.

10 Cournot reading and CNMC enforcement context

10.1 Cournot reading of the cap-padded ladder

A near-constant DA bid price ladder at the cap (GN at $\sim$ 840 EUR/MWh in Table 15) combined with all the adjustment on the *quantity* side — the unit either clears DA or it does not, but its bid price hardly varies — is the empirical signature of a **Cournot strategy on the quantity margin**. In pure economic terms this is equivalent to direct price-raising in DA, but the regulatorily defensible posture for the firm is "my offer price has not changed; my bids follow my published ladder." The redispatch channel (Fase I pay-as-bid) is then where the per-MWh rent is captured.

10.2 Historical CNMC sanction precedent

The empirical pattern documented in §§6–8 has a direct precedent in CNMC enforcement. The resolutions for the relevant cases are archived in the project's regulation document folder.

Table 20: CNMC sanction resolutions relevant to the practice. "LSE" = Ley del Sector Eléctrico 24/2013. Sources: CNMC resolution PDFs archived in the project regulation folder.

Year	Firm / Plant	Expediente	LSE article	Sanction
2015	Iberdrola hydro	SNC-DE-046-14	64 muy grave (manipulation)	25.0 M EUR
2019	**Naturgy CCGT**†	SNC-DE-175-17	65.34 grave (abnormal DA offers)	~7 M EUR
2019	Endesa Besós 3	SNC-DE-174-17	65.34 grave (abnormal DA offers)	6.6 M EUR
2023	Naturgy Sabón 3 (Galicia)	—	65.33 grave (RT abuse)	confidential
2023	Engie Castelnou	SNC-DE-152-22	grave (availability)	—
2023	Ignis Escatrón 2	SNC-DE-151-22	grave (availability)	—

† Plants in scope: Besós 4, PBCN 1/2, Sagunto 1/2/3, Málaga 1, SROQ 1 — exactly the Cataluña / Levante / Sur zonal map (§8). CNMC-estimated minimum benefit 13.0 M EUR over 4 months.

Reading. The 2019 Naturgy sanction (SNC-DE-175-17) is the smoking-gun precedent: same plants, same zones, same mechanism (high DA → unit not despatched in DA → unit systematically programmed at higher RT prices) over October 2016 – January 2017. The 2024–26 evidence in this Part B is the same practice continuing post-*operación reforzada*, now at fleet-wide scale, with the within-hour granularity reform as an additional tool. The 2023 Sabón 3 sanction extends the geography to Galicia (the GE / GN duopoly per §8).

10.3 Blackout-cohort expedientes (separate channel)

In April 2026 CNMC opened a parallel set of ∼ 56 expedientes (IB 22, GE 17, GN 5, others) under Art 65.8 of the LSE, for **voltage-control violations** on April 28, 2025 (the day of the blackout). This is a *distinct regulatory channel*: it concerns failure to provide reactive support during the blackout sequence, not auction-shaping in DA. We flag it for completeness; the two channels are not conflated here.

10.4 The blackout-attribution question: ENTSO-E vs CNMC

A separate but related question is whether the March 2025 intraday-granularity reform (ID15) contributed to the April 2025 blackout itself — which would, indirectly, attribute the post-blackout reforzada cost rise to the same reform. The question is out of scope for this thesis (it requires electrical-engineering knowledge we do not have), but the two official investigations of the incident take different positions and the contrast is worth recording. *Preliminary results* §5 quotes the full CNMC excerpt (in English translation with the original Spanish in footnote); we summarise the contrast here.

- **ENTSO-E Expert Panel final report (`ICS` investigation, March 2026)** does not list ID15 as a contributing factor. Its root-cause tree points to voltage- and reactive-power-control failures: renewable plants operating at fixed power factor, manually-operated shunt reactors, divergent overvoltage protection settings, converter-driven instability, absence of power-system stabilisers on several large units. The market-microstructure layer does not enter.

- **CNMC report on the incident** (`PRO/CNMC/001/26`, March 2026), by contrast, *anecdotally* flags the move to quarter-hourly trading as one contributor to the abrupt voltage variations observed in the months around the incident: the report relays the system operator's reading that quarter-hourly intraday trading (in force from 2025-03-19, one month before the blackout) and quarter-hourly day-ahead trading (from 2025-10-01) increased the frequency of large within-hour production swings, including transitions between negative and positive prices that drive sharp volume changes in certain zones.

The two investigations are at different epistemic standing — ENTSO-E is the authoritative technical investigation with full TSO access; CNMC's flag is a single paragraph relaying the system operator's reading without independent technical substantiation. We do not adjudicate. The preliminary memo treats the CNMC flag as an *anecdotal pointer to a research gap*: the ENTSO-E report identifies the proximate technical triggers but does not engage with whether the market-microstructure reforms contributed to the macro-conditions (frequent within-hour swings, converter-driven volatility under fixed-power-factor RES dispatch) that made the system fragile enough for those triggers to cascade. This is the kind of question that would require a joint microstructure-and-electrical-engineering analysis — a natural extension well beyond this thesis but worth flagging as the strongest implication of the empirical work documented here.